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Adversarial Challenges in Network Intrusion Detection Systems: Research Insights and Future Prospects

SABRINE ENNAJI, Sapienza University of Rome, Italy
FABIO DE GASPARI, Sapienza University of Rome, Italy
DORJAN HITAJ, Sapienza University of Rome, Italy
ALICIA K BIDI, Sapienza University of Rome, Italy
LUIGI VINCENZO MANCINI, Sapienza University of Rome, Italy

Machine learning has brought significant advances in cybersecurity, particularly in the development of Intrusion Detection Systems (IDS). These improvements are mainly attributed to the ability of machine learning algorithms to identify complex relationships between features and effectively generalize to unseen data. Deep neural networks, in particular, contributed to this progress by enabling the analysis of large amounts of training data, significantly enhancing detection performance. However, machine learning models remain vulnerable to adversarial attacks, where carefully crafted input data can mislead the model into making incorrect predictions. While adversarial threats in unstructured data, such as images and text, have been extensively studied, their impact on structured data like network traffic is less explored.

This survey aims to address this gap by providing a comprehensive review of machine learning-based Network Intrusion Detection Systems (NIDS) and thoroughly analyzing their susceptibility to adversarial attacks. We critically examine existing research in NIDS, highlighting key trends, strengths, and limitations, while identifying areas that require further exploration. Additionally, we discuss emerging challenges in the field and offer insights for the development of more robust and resilient NIDS. In summary, this paper enhances the understanding of adversarial attacks and defenses in NIDS and guide future research in improving the robustness of machine learning models in cybersecurity applications.

Additional Key Words and Phrases: Network intrusion detection systems, machine learning, cybersecurity, deep learning, adversarial attacks

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1 INTRODUCTION

The widespread deployment of internet-connected devices in our everyday lives has significantly impacted how we communicate, work, learn, and interact with the world. As society progresses in the information age, the Internet has evolved from a mere convenience to an indispensable necessity, playing a vital role in how our society functions, businesses operate, and people connect. However, this high level of interconnectivity also creates

Authors' addresses: Sabrine Ennaji, ennaji@di.uniroma1.it, Sapienza University of Rome, Rome, Italy; Fabio De Gaspari, degaspari@di.uniroma1.it, Sapienza University of Rome, Rome, Italy; Dorjan Hitaj, hitaj.d@di.uniroma1.it, Sapienza University of Rome, Rome, Italy; Alicia K bidi, kbidi@di.uniroma1.it, Sapienza University of Rome, Rome, Italy; Luigi Vincenzo Mancini, mancini@di.uniroma1.it, Sapienza University of Rome, Rome, Italy.

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網路入侵偵測系統中的對抗性挑戰：研究洞見與未來展望

SABRINE ENNAJI，羅馬大學，義大利
FABIO DE GASPARI，羅馬大學，義大利
DORJAN HITAJ，羅馬大學，義大利
ALICIA K BIDI，羅馬大學，義大利
LUIGI VINCENZO MANCINI，羅馬大學，義大利

機器學習為網路安全帶來了顯著的進步，特別是在入侵偵測系統（IDS）的開發上。這些進步主要歸功於機器學習演算法能夠識別特徵間的複雜關係，並有效地泛化到未見過的數據。深度神經網路，特別地，透過能夠分析大量訓練數據，顯著提升了偵測性能，為此進步貢獻了力量。然而，機器學習模型仍然容易受到對抗性攻擊，即精心設計的輸入數據會誤導模型做出錯誤的預測。雖然對抗性威脅在圖像和文本等非結構化數據上已被廣泛研究，但它們對網路流量等結構化數據的影響則較少被探討。本調查旨在填補這個空白，透過提供機器學習基礎的網路入侵偵測系統（NIDS）的全面回顧，並深入分析其對抗性攻擊的易受性。我們嚴謹地檢視了NIDS的現有研究，突顯了關鍵趨勢、優點和限制，同時識別了需要進一步探討的領域。此外，我們討論了該領域的新興挑戰，並為開發更穩固和更具韌性的NIDS提供見解。總之，本文增進了對NIDS中對抗性攻擊和防禦的理解，並為未來研究改善機器學習模型在網路安全應用中的穩健性提供指導。

額外關鍵詞與短語：網路入侵偵測系統、機器學習、網路安全、深度學習、對抗性攻擊

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1 緒論

我們日常生活中網路連接設備的廣泛部署，已顯著影響了我們的溝通、工作、學習以及與世界互動的方式。隨著社會在資訊時代進步，互聯網已從一種僅僅是便利的工具演變為不可或缺的必需品，在社會運作、企業運營以及人際連接中發揮著至關重要的作用。然而，這種高度互連性也創造了

作者地址：Sabrine Ennaji, ennaji@di.uniroma1.it, 羅馬大學Sapienza校區, 羅馬, 意大利；Fabio De Gaspari, degaspari@di.uniroma1.it, 羅馬大學Sapienza校區, 羅馬, 意大利；Dorjan Hitaj, hitaj.d@di.uniroma1.it, 羅馬大學Sapienza校區, 羅馬, 意大利；Alicia K bidi, kbidi@di.uniroma1.it, 羅馬大學Sapienza校區, 羅馬, 意大利；Luigi Vincenzo Mancini, mancini@di.uniroma1.it, 羅馬大學Sapienza校區, 羅馬, 意大利.

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opportunities for adversaries to exploit vulnerabilities in internet-facing services and devices for different purposes, including economic, political, or personal gain. Some attacks exploit network protocol vulnerabilities [16, 22], others focus on extracting and exploiting information through monitoring and sniffing network traffic [37, 148], while others merely use the network as a medium to reach and misuse vulnerable systems and services [12, 48].

Given the threat that cyberattacks present, improving the security of networks and computer systems has been a growing concern. Modern defenses typically rely on defense-in-depth approaches involving a variety of different strategies, such as firewalls [56], cryptographic techniques [97], host-based malware detection [52], autonomous defense systems [43, 139], moving-target defense [75, 112], and Intrusion Detection Systems (IDS) [111]. Network Intrusion Detection Systems (NIDS), in particular, play a significant role in detecting various types of anomalous behavior due to their flexibility and ability to detect potential attacks before they reach a host by observing network traffic [24]. They are known to be efficient and have good potential to provide external and internal protection across services, devices, and architectures. In contrast to host-based IDS, NIDS do not need to run on each device that requires protection, but only require access to relevant network traffic. Despite these advantages, traditional NIDS based on expert knowledge require continuous updating to remain effective against evolving threats, as well as expert supervision to define detection rules. Over time, this can lead to performance degradation, reduced capability to detect novel attacks, and increased false positive rates. To address this gap, Machine Learning (ML) techniques and, more recently, Deep Learning (DL) have been increasingly integrated into NIDS to reduce the reliance on expert knowledge, improve generalization performance, and decrease the costs and complexity associated with updates [36]. ML, and particularly Deep Neural Network (DNN), has revolutionized several domains due to the ability to extract valuable features from data and make accurate predictions. This is demonstrated by its success in areas like image classification [35, 64], natural language processing [81, 113], speech recognition [57, 65], data generation [66, 80], cybersecurity [68–70], forensics analysis [40, 42], and even applications related to the COVID-19 pandemic [89, 96].

Critically, while extremely effective in controlled environments, ML techniques are susceptible to various attacks when exposed to adversarial settings [18, 103]. Many attacks against ML have surfaced over the years, ranging from training-time attacks (e.g., backdooring, poisoning [38, 140]) to inference-time evasion attacks (e.g., adversarial examples [55, 155]). Evasion attacks, in particular, present a fundamental threat to NIDS, as attackers possess considerable flexibility in modifying their traffic patterns to evade detection. Adversarial examples fall in this category of attacks, manipulating legitimate traffic data with subtle modifications to deceive the ML model and achieve an undesired output (incorrect classification). While adversarial examples were originally designed for unstructured data in the continuous domain [55], the significant risk they create spurred research investigating their applicability to other domains, such as network traffic, in an effort to respond to the question:

Can adversarial attacks also threaten structured data?

Recent studies [39, 86, 123, 131, 146] demonstrate the applicability of evasion attacks and adversarial examples to structured data, introducing a significant vulnerability to cybersecurity applications, including ML-NIDS. However, existing research on NIDS design and evaluation often fails to adequately address this critical threat. This gap in knowledge motivates this work in three ways:

- **Raise awareness:** Currently, a limited number of studies about ML-NIDS include adversarial attacks in their threat models. Due to the difficulties in adhering to the constraints and interdependencies among features in structured data, these attacks are frequently overlooked. Furthermore, adaptive attacks designed to evade detection techniques remain only partially addressed in NIDS research. Given this potential risk to ML-NIDS, this paper aims to increase the awareness of these attacks in the NIDS community.
- **Provide a clear point of reference:** This developing field lacks comprehensive surveys on adversarial examples against NIDS. To the best of our knowledge, there is no existing survey that provides a full taxonomy of ML-NIDS, explores the interplay between adversarial examples and feature interdependencies

給予敵對勢力利用互聯網面向服務與設備中漏洞的機會，以達成經濟、政治或個人利益等不同目的。部分攻擊利用網路協議漏洞 [16, 22]，其他則專注於透過監控和嗅探網路流量來萃取和利用資訊 [37, 148]，而另一些則僅將網路視為達成和濫用脆弱系統與服務的媒介 [12, 48]。

考量到網路攻擊所造成的威脅，提升網路與電腦系統的安全性已成為日益關注的議題。現代防禦通常依賴多層式防禦策略，包含各種不同的方法，例如防火牆 [56]，加密技術 [97]，主機基礎的惡意軟體偵測 [52]，自動防禦系統 [43, 139]，移動目標防禦 [75, 112]，與入侵偵測系統 (IDS) [111]。網路入侵偵測系統 (NIDS)，特別是，由於其靈活性與能在攻擊觸及主機前偵測潛在攻擊的能力（透過觀察網路流量 [24]），在偵測各種異常行為方面扮演重要角色。它們被證明效率很高，並且在服務、設備和架構方面有良好的潛力提供外部和內部保護。與主機基礎的 IDS 相比，NIDS 不需要在每個需要保護的設備上運行，只需要有權限存取相關的網路流量。儘管有這些優勢，基於專家知識的傳統 NIDS 需要持續更新才能對演變的威脅保持有效，以及需要專家監督來定義偵測規則。隨著時間的推移，這可能導致性能下降、偵測新式攻擊的能力降低，以及偵測錯誤率增加。為了彌補這個差距，機器學習 (ML) 技術，以及較新的深度學習 (DL)，已經越來越多地整合到 NIDS 中，以減少對專家知識的依賴，提升泛化性能，並降低與更新相關的成本和複雜性 [36]。ML，特別是深度神經網路 (DNN)，由於能夠從數據中提取有價值的特徵並做出準確預測，已經革命化了幾個領域。這一點在其在圖像分類 [35, 64]，自然語言處理 [81, 113]，語音辨識 [57, 65]，數據生成 [66, 80]，網路安全 [68–70]，經濟分析 [40, 42]，甚至與 COVID-19 疫情相關的應用 [89, 96] 等領域的成功中得到證明。

至關重要地，雖然機器學習技術在受控環境中極其有效，但在遭遇敵對設定時，它們會受到各種攻擊的威脅 [18, 103]。多年來，針對機器學習的攻擊層出不窮，從訓練時攻擊（例如，植入後門、毒化 [38, 140]）到推理時的迴避攻擊（例如，敵對樣本 [55, 155]）。特別是迴避攻擊，對於網路入侵偵測系統（NIDS）構成基本威脅，因為攻擊者擁有相當大的彈性來修改其流量模式以迴避偵測。敵對樣本屬於這類攻擊，通過對合法流量數據進行微妙的修改來欺騙機器學習模型，並達成不願見的輸出（錯誤分類）。雖然敵對樣本最初是為連續域中的非結構化數據設計的 [55]，但它們造成的顯著風險促使研究人員調查其對其他領域（例如，網路流量）的適用性，以回應這個問題：

敵對攻擊也能威脅結構化數據嗎？

近年來的研究 [39, 86, 123, 131, 146] 已證明迴避攻擊與對抗性範例對於結構化資料的適用性，為網路安全應用帶來顯著的弱點，包括ML-NIDS。然而，現有針對NIDS設計與評估的研究，往往未能充分應對這項嚴峻的威脅。這項知識上的落差，從三個面向激勵了本研究：

- 提高意識：目前，關於機器學習-NIDS的研究中，只有有限數量的研究將敵對攻擊納入其威脅模型。由於在遵守結構化數據中特徵之間的限制和相互依存關係存在困難，這些攻擊經常被忽略。此外，設計用於迴避偵測技術的自適應攻擊在NIDS研究中僅僅得到部分處理。考慮到這對於機器學習-NIDS的潛在風險，本文旨在提高NIDS社區對這些攻擊的意識。
- 提供明確的參考點：這個發展中的領域對於針對NIDS的對抗範例缺乏全面的調查。根據我們所知，目前沒有現有的調查能夠提供ML-NIDS的完整分類法，也沒有調查探討對抗範例與特徵間的相互依存關係

within NIDS, and summarizes existing knowledge on evasion attacks and defenses for NIDS. We aim to bridge this gap and provide a new reference point for NIDS research, covering all aspects of robust network intrusion detection.
- **Establish best practices**: Existing studies on ML-NIDS use different datasets, evaluation metrics, and settings, which makes it difficult to compare findings and identify strengths and weaknesses within proposed solutions. This paper proposes a set of guidelines and best practices for consistent ML-NIDS evaluation, emphasizing the importance of testing against adaptive attacks.

CONTRIBUTIONS. This survey provides a comprehensive, analytical review of ML-based NIDS and recent practical applications of adversarial attacks against NIDS. In summary, we make the following contributions:
(1) We review recent advancements in ML-based NIDS, providing a thorough taxonomy and analysis of existing approaches.
(2) We compare and contrast different adversarial attacks against ML models, comprehensively discussing how they affect ML-based NIDS.
(3) We discuss and critically review existing defenses against adversarial attacks on NIDS.
(4) We provide an extensive analysis of lessons learned, novel insights on adversarial attacks on NIDS, and good practices and guidelines for future research.

ORGANIZATION OF THE SURVEY. Figure 1 outlines a schematic representation of this paper's structure, which is organized as follows. Section 2 reviews existing IDS surveys and highlights our contributions. Section 3 provides general background on the evolution of NIDS, ML, DL, adversarial attacks, and their countermeasures. A comprehensive literature review of ML-NIDS is presented in Section 4. Section 5 critically reviews literature on adversarial attacks against ML-NIDS, while Section 6 expands on this to review possible defenses. Section 7 summarizes identified challenges and presents lessons learned, insights, and future research guidelines. Finally, Section 8 concludes our work.

2 COMPARISON TO OTHER SURVEYS

Adversarial attacks against NIDS is an emerging research area lying at the intersection of cybersecurity and ML. While extensive surveys exist on NIDS, there is a lack of comprehensive analysis specifically focused on adversarial attacks against them. Existing studies fall into three categories: (1) surveys on ML and DL-based NIDS, (2) surveys on adversarial attacks and defenses in the continuous domain, and (3) surveys on adversarial attacks on malware detection. In the following sections, we critically analyze related works, identifying their limitations and highlighting the innovative aspects of our survey.

2.1 Surveys on Traditional Machine Learning and Deep Learning Applications to NIDS

Several surveys analyze the application of traditional ML and DL to NIDS. A foundational work in this area is by Buczak et al. [21]. They conducted a comprehensive literature study on ML for network intrusion detection around 2016. Their work presents a complete taxonomy of existing NIDS solutions and provides tutorial descriptions of each ML applied therein. In addition to an overview and description of commonly used datasets to train and test NIDS. Building upon this foundation, Chaabouni et al. [29] analyzed ML-NIDS techniques for securing Internet of Things (IoT) devices. Similarly to Buczak et al., they propose a thorough taxonomy of NIDS, refining it for the IoT context to address domain-specific problems and challenges. Ahmad et al. [2] furthered this progress by defining an updated taxonomy of ML-NIDS and critically reviewing existing techniques, highlighting challenges and proposing new research directions. Many more general surveys on NIDS exist [47, 125, 137], providing similar contributions and including the most recent ML-NIDS approaches proposed in the literature. However, while these references are valuable for new researchers in the field, they all share a crucial limitation: they fail to consider the problem of adversarial attacks to ML-NIDS and related countermeasures.

針對NIDS內部，並總結現有關於迴避攻擊與NIDS防禦的知識。我們旨在填補這項缺口，為NIDS研究提供新的參考基準，涵蓋所有關於強健網路入侵偵測的面向。

- 建立最佳實踐：現有針對ML-NIDS的研究使用不同的數據集、評估指標和設定，這使得比較研究結果並識別所提出解決方案中的優缺點變得困難。本文提出一套用於ML-NIDS一致性評估的指導原則和最佳實踐，強調了對抗適應性攻擊測試的重要性。

貢獻。本調查對基於ML的NIDS進行了全面、分析的回顧，並總結了針對NIDS的對抗攻擊的最新實際應用。總之，我們做出了以下貢獻：
(1) 我們回顧了基於ML的NIDS的最新進展，提供了一個全面的分類和對現有方法的分析。
(2) 我們比較和對比了針對ML模型的各種對抗攻擊，全面討論了它們如何影響基於ML的NIDS。
(3) 我們討論並批判性地回顧了針對NIDS上對抗攻擊的現有防禦措施。
(4) 我們提供對經驗教訓的深入分析、對NIDS（網路入侵偵測系統）中對抗性攻擊的新見解，以及未來研究的好實踐和指導方針。
調查的組織。圖1概述了本文結構的示意圖，其組織方式如下。第2節回顧了現有的IDS（入侵偵測系統）調查並強調了我們的貢獻。第3節提供了關於NIDS、機器學習（ML）、深度學習（DL）、對抗性攻擊及其對策的背景資訊。第4節呈現了對ML-NIDS的全面文獻回顧。第5節批判性地回顧了針對ML-NIDS的對抗性攻擊的文獻，而第6節則擴展這一內容，審視可能的防禦措施。第7節總結了已識別的挑戰並提出經驗教訓、見解和未來研究指導方針。最後，第8節總結了我們的工作。

2 與其他調查的比較

針對NIDS的對抗性攻擊是一個處於網路安全性和機器學習交叉點的新興研究領域。雖然存在關於NIDS的廣泛調查，但缺乏專門針對其對抗性攻擊的全面分析。現有研究可分為三類：(1) 基於ML（機器學習）和DL（深度學習）的NIDS調查，(2) 連續域中對抗性攻擊和防禦的調查，以及(3) 針對惡意軟體偵測的對抗性攻擊調查。在以下幾節中，我們批判性地分析相關研究，識別其限制並強調我們調查的創新之處。

2.1 執行傳統機器學習與深度學習應用於NIDS的調查

數份調查分析傳統機器學習（ML）與深度學習（DL）在網路入侵偵測系統（NIDS）的應用。此領域的基礎性工作是Buczaket等人 [21]的研究。他們在2016年左右進行了一項全面的機器學習（ML）網路入侵偵測文獻研究。他們的工作提出了一個完整的現有NIDS解決方案分類法，並提供每種應用於其中的機器學習的教學描述。此外，還概述和描述了用於訓練和測試NIDS的常用數據集。在這個基礎上，Chaabouni等人 [29] 分析了用於保護互聯網物聯網（IoT）設備的ML-NIDS技術。與Buczak等人相似，他們提出了一個全面的NIDS分類法，並針對IoT語境進行細化，以解決特定領域的問題和挑戰。Ahmad等人 [2] 通過定義一個更新的ML-NIDS分類法，並批判性地評論現有技術，突顯挑戰並提出新的研究方向，進一步推進了這項進展。還有許多關於NIDS的普遍調查 [47, 125, 137], 提供了類似的貢獻，並包括文獻中提出的最新ML-NIDS方法。然而，雖然這些參考對該領域的新研究員來說很有價值，但它們都有一個關鍵的局限性：它們未能考慮針對ML-NIDS的對抗性攻擊以及相關的對抗措施。

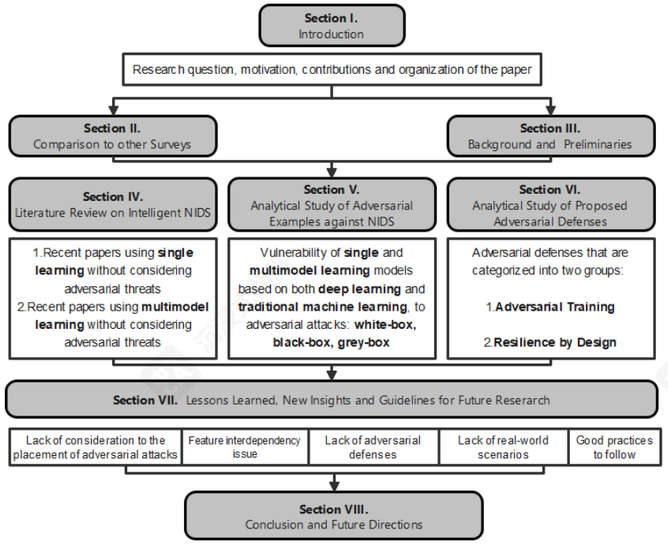


Fig. 1. Structure of the Presented Survey

2.2 Surveys on Adversarial Attacks and Defenses in the Continuous Domain

Adversarial attacks against ML models in the continuous domain are a well-known challenge, both for traditional ML [18] and DNN [149, 158]. Barreno et al. [18] propose one of the first comprehensive analyses of the security of traditional ML techniques against adversarial attacks. In their work, the authors define a complete taxonomy of adversarial attacks against traditional ML algorithms and study the effectiveness of such attacks against practical applications such as spam filtering. In [158], Zhou et al. survey existing adversarial attacks against DNN, both at training and inference time. The authors provide a comprehensive classification for such attacks and define a lifecycle for adversarial attacks and defenses. Xiong et al. [149] propose a survey on the effects of different properties of training data on the adversarial robustness of DNN. The authors analyze several published works and propose a set of guidelines and techniques for improving adversarial robustness through data manipulation.

This line of surveys provides an invaluable resource for understanding all the existing constructions for adversarial attacks and defenses. However, all of them focus on the continuous domain (mainly images) and are not applicable to the context of NIDS, where complex interdependencies exist between features and constraints on feature values. Moreover, they typically do not consider challenges specific to cybersecurity and NIDS.

2.3 Surveys on Adversarial Attacks in Malware Detection

General Malware Detection: Recently, researchers started studying the effects of adversarial examples in the area of malware detection. Gardiner et al. [51] studied the security of traditional ML techniques used for detecting malware's command and control channels. Their work systematically categorized existing command and control detection methods and analyzed the various attacks that could exploit these detectors. In [116],

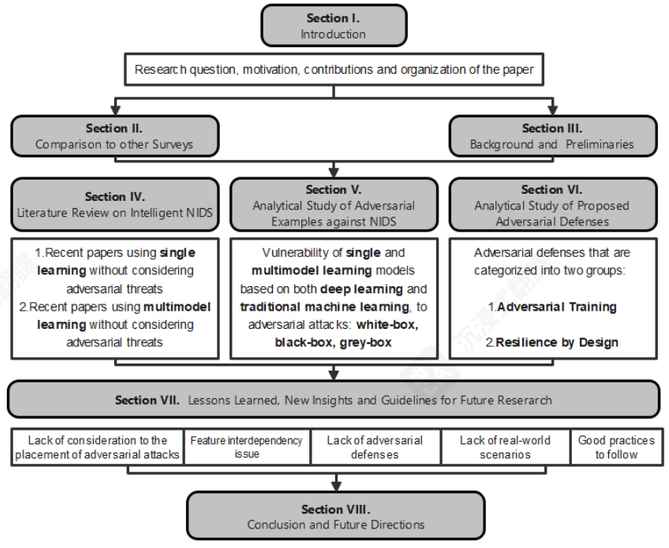


圖1。所提出調查的結構

2.2 關於連續域中的對抗攻擊與防禦的調查

針對連續域中機器學習模型的對抗性攻擊是一個众所周知的挑戰，既針對傳統機器學習 [18] ，也針對深度神經網絡（DNN） [149, 158]。Barreno等人 [18] 提出了第一個全面的傳統機器學習技術對抗性攻擊安全性分析之一。在他们的工作中，作者定義了一個完整的針對傳統機器學習算法的對抗性攻擊分類法，並研究了這種攻擊對垃圾郵件過濾等實際應用程序的有效性。在 [158], Zhou等人調查了針對DNN的現有對抗性攻擊，包括訓練和推理時間。作者為這種攻擊提供了全面的分類，並定義了對抗性攻擊和防禦的生命週期。Xiong等人 [149] 提出了一項調查，探討了訓練數據的不同屬性對DNN對抗性強度的影響。作者分析了幾篇已發表的著作，並提出了一組指導原則和技術，以通過數據操縱來提高對抗性強度。

這些調查為理解所有現有的對抗攻擊與防禦構建提供了寶貴的資源。然而，它們全都集中在連續域（主要是圖像）上，不適用於NIDS的背景，在NIDS中，特徵之間存在複雜的相互依存關係，並且對特徵值施加了限制。此外，它們通常不考慮特定於網絡入侵檢測和網絡安全挑戰的挑戰。

2.3 關於惡意軟體檢測中的對抗攻擊的調查

一般惡意軟體偵測：最近，研究人員開始研究惡意軟體偵測領域中對抗性範例的影響。Gardiner 等人. [51] 研究了用於偵測惡意軟體指令與控制頻道的傳統機器學習技術的安全性。他們的工作系統性地分類了現有的指令與控制偵測方法，並分析了可以利用這些偵測器的各種攻擊。在 [116],

Table 1. Contributions of this survey in comparison to most the relevant previous works

Reference	Year	ML NIDS Taxonomy	Feature Interdependency	Placement of Adv. Ex.	Real-World Challenges	NIDS Adv. Ex. Defenses	Lessons learned for NIDS
Ibitoye et al. [72]	2019	✗	✓	✗	✗	✓	✓
Olowononi et al. [110]	2020	✓	✗	✗	✗	✗	✗
Martins et al. [95]	2020	✗	✓	✗	✓	✗	✗
Alotaibi et al. [10]	2023	✗	✗	✗	✓	✗	✗
Pinto et al. [120]	2023	✗	✓	✗	✓	✗	✗
He et al. [63]	2023	✓	✓	✗	✗	✓	✓
Our survey	**2024**	✓	✓	✓	✓	✓	✓

Park et al. provide a comprehensive survey of practical adversarial attacks against malware classifiers. Their paper explores various adversarial example-generation techniques, developing a framework to categorize them, and analyzes their strengths and weaknesses. More recently, Rosenberg et al. [124] propose in-depth survey on adversarial machine learning attacks and defenses specific to the cybersecurity field. Their study establishes a characterization of adversarial attacks based on three key aspects: lifecycle, adversarial goals, and attacker capabilities. Furthermore, they categorize different defense methods employed in cybersecurity applications. These and other similar surveys (e.g., [3], [84]) respond to the broader challenges of adversarial attacks on ML and robust malware detection. However, they lack a comprehensive analysis specific to NIDS and often focus on a particular area in adversarial attacks, neglecting the unique issues that arise within NIDS.

Finally, the analysis of potential defenses, when present, tends to be general and not specific to the context of malware detection.

Intrusion Detection Systems: In [110], Olowononi et al. review ML-based defenses for Networked Cyber-Physical Systems (CPS), examining recent intrusion detection techniques in CPS that leverage ML, with a focus on their robustness against adversarial attacks. Their study provides a full taxonomy of traditional ML and DNN intrusion detection methods for CPS and identifies promising future research directions. While the authors investigate the challenge of developing robust ML applications for intrusion detection, their focus lies in the distinct domain of CPS, separate from the area addressed by our survey.

Our paper provides a comprehensive taxonomy and in-depth analysis of ML-NIDS, identifying and exploring the unique challenges within this domain. Additionally, it explores methods specifically designed for adversarial attacks against NIDS, in contrast to [110], which focuses on general adversarial attacks. In [95], Martins et al. survey applications of adversarial machine learning in intrusion and malware detection; they categorize general attack strategies and defenses before applying them to intrusion detection. Notably, their study lacks a general taxonomy of NIDS. Our survey distinguishes itself by presenting a detailed categorization of both ML-NIDS and the specific adversarial attacks that target them. We further present a thorough analysis of defenses against adversarial attacks for NIDS, which is an absent aspect in [95]. Moreover, we discuss new insights, valuable lessons learned, and promising research directions in this field. While in another thorough summary of several adversarial attacks and defenses for ML-NIDS explored by Ibitoye et al. [72], it does not provide a specific taxonomy of NIDS, but covers a wide variety of detectors such as spam detection, phishing detection, and general malware detection. Moreover, it does not consider the applicability of the discussed techniques. The presented defenses are also general and not specific to NIDS. Additionally, in [10] an analysis of adversarial attacks with related defenses for ML-IDS is provided, but without discussing existing gaps in this area. A comprehensive taxonomy of ML-NIDS is not considered as well. Similarly, Pinto et al. [120] survey applications of ML to intrusion detection to protect critical infrastructure. A high-level categorization of ML-based defenses is introduced, but the challenge

表1. 本調查與相關先前工作的貢獻比較

參考資料	Year	ML NIDS 分類法	特性相互依存	副詞位置	實際應用挑戰	NIDS優勢防禦範例	NIDS所學到的教訓
Ibitoye等人 [72]	2019	✗	✓	✗	✗	✓	✓
Olowononi等人. [110]	2020	✓	✗	✗	✗	✗	✗
Martins等人. [95]	2020	✗	✓	✗	✓	✗	✗
Alotaibi等人 [10]	2023	✗	✗	✗	✓	✗	✗
Pinto等人. [120]	2023	✗	✓	✗	✓	✗	✗
He 等人. [63]	2023	✓	✓	✗	✗	✓	✓
我們的調查	**2024**	✓	✓	✓	✓	✓	✓

Park 等人提供了一份關於針對惡意軟體分類器的實用對抗性攻擊的全面調查。他們的論文探討了各種對抗性範例生成技術，開發了一個用於分類它們的框架，並分析了它們的優缺點。最近，Rosenberg 等人. [124] 針對網路安全領域的對抗性機器學習攻擊與防禦提出了一份深入調查。他們的研究基於三個關鍵方面——生命週期、對抗性目標和攻擊者能力——建立了一種對抗性攻擊的描述。此外，他們還分類了網路安全應用中使用的不同防禦方法。這些以及其他類似的調查（例如， [3], [84]）回應了對抗性攻擊對機器學習和穩固惡意軟體偵測提出的更廣泛挑戰。然而，它們缺乏針對NIDS的全面分析，並且通常只關注對抗性攻擊的特定領域，忽略了NIDS中出現的獨特問題。

最後，潛在防禦的分析，當存在時，往往較為概括，並不針對惡意軟體偵測的具體情境。

入侵偵測系統：在 [110], Olowononi 等人的研究中，他們檢視了基於機器學習的網絡化 cyber-physical systems (CPS) 防禦措施，探討了 CPS 中利用機器學習的最近入侵偵測技術，並聚焦於其對抗敵對攻擊的穩健性。他們的研究為 CPS 的傳統機器學習和深度神經網路入侵偵測方法提供了完整的分類學，並確定了有前景的未來研究方向。雖然作者們研究了開發用於入侵偵測的穩健機器學習應用的挑戰，但他們的焦點位於獨特的 CPS 領域，與我們調查所處理的領域不同。

我們的論文提供了一個全面的分類法以及對 ML-NIDS 的深入分析，識別並探討了這個領域中的獨特挑戰。此外，它探討了針對 NIDS 的對抗性攻擊的特定方法，與 [110], 相比，後者專注於一般對抗性攻擊。在 [95], Martins 等人的調查中，探討了對抗性機器學習在入侵和惡意軟體檢測中的應用；他們在將其應用於入侵檢測之前，先分類了一般的攻擊策略和防禦措施。值得注意的是，他們的研究缺乏對 NIDS 的通用分類法。我們的調查通過呈現對 ML-NIDS 和針對它們的特定對抗性攻擊的詳細分類，使其區別於其他研究。我們進一步提出了一個對 NIDS 的對抗性攻擊防禦的全面分析，這是 [95] 中缺失的一環。此外，我們討論了這個領域的新見解、寶貴的經驗教訓和有前景的研究方向。雖然在 I bitoye 等人 [72], 的另一份對 ML-NIDS 的對抗性攻擊和防禦的全面總結中，它沒有提供 NIDS 的特定分類法，但涵蓋了廣泛的檢測器，例如垃圾郵件檢測、網絡釣魚檢測和一般惡意軟體檢測。此外，它沒有考慮所討論技術的適用性。所提出的防禦措施也是一般的，並不針對 NIDS。此外，在 [10] 中提供了一個對 ML-IDS 的對抗性攻擊及其相關防禦的分析，但沒有討論這個領域的現有差距。ML-NIDS 的全面分類法也不被考慮。類似地，Pinto 等人 [120] 調查了應用 ML 進行入侵檢測以保護關鍵基礎設施的應用。引進了基於 ML 的防禦措施的高層次分類，但對抗性示例的挑戰

of adversarial examples is not thoroughly addressed. Finally, He et al. [63] propose a comprehensive analysis of existing works in ML-NIDS. They present a taxonomy of NIDS, a categorization of adversarial attacks and defenses against NIDS. In addition to an analysis of future research directions for this field. Compared to [63], not only does our survey include additional recent papers in the literature, but we also critically review existing techniques, highlighting limitations and potential challenges to adoption. Moreover, we discuss new observations and correlations between the placement of adversarial examples within the data manifold [90] and the resilience to adversarial attacks.

2.4 Comparison Summary

In summary, compared to related works, our survey is distinguished by comprehensive and methodical approach. We propose a complete taxonomy of ML-NIDS, adversarial attacks targeting them, and potential defenses. This analysis goes beyond simply listing approaches; we evaluate their strengths and weaknesses, providing summaries to guide practitioners. We further explore the real-world applicability of these attacks with their defenses in the NIDS context. Notably, our survey shed light on new insights on the existence and placement of adversarial examples and the interdependencies between malware features in NIDS. By analyzing these aspects and conducting a comprehensive review, we extract valuable lessons learned and propose a set of guidelines for developing robust ML-NIDS. Table 1 summarizes the comparison of our survey to the most relevant related works and highlights significant gaps in existing surveys on adversarial examples against ML-NIDS. Surveys serve an indispensable purpose in consolidating knowledge and identifying research directions, which supports the advancement of a discipline. We believe our work offers detailed analysis and novel insights, making it a valuable resource for both new and experienced NIDS practitioners.

3 BACKGROUND

This section provides background information regarding intrusion detection systems, deep neural networks, common adversarial attacks on learning systems, and existing countermeasures.

3.1 Intrusion Detection Systems

Intrusion Detection Systems include network traffic monitoring (NIDS) and system behavior analysis (host-based IDS) to detect cyberattacks [6]. They work in conjunction with other security mechanisms, including firewalls, and use various detection techniques categorized into two main types:

- **Signature-based detection** uses information collected from previous attacks and known vulnerabilities to create a characteristic and recognizable signature. As attackers continuously update their attack patterns and methods, signatures become quickly obsolete and need to be regularly updated to remain effective [71]. Given its reliance on known attack patterns, signature-based detection typically offers high precision with low false positive rates but fails to detect new, unknown attacks.
- **Anomaly-based detection** models the typical behavior of systems and networks and detects deviations from such behavior as potential attacks. The main advantage of this approach lies in its ability to identify both known and unknown attacks, as long as they cause a detectable behavioral shift. Furthermore, once the typical behavior of a system/network is computed, anomaly-based IDS do not need continuous updates. However, anomaly-based detection generally results in a higher rate of false positives compared to signature-based methods, making it more time-consuming to handle false alerts.

3.2 Intelligent Intrusion Detection Systems

The growing complexity of cyberattacks led to significant advancements in IDS. In recent years, there has been a significant push towards integration of ML techniques in IDS, greatly enhancing detection capabilities [127].

沒有得到充分說明。最後，He 等人 [63] 提出了一個對 ML-NIDS 現有工作的全面分析。他們提出了一個 NIDS 的分類法，以及對抗性攻擊和防禦措施的分類。此外，還進行了對這個領域未來研究方向的分析。與 [63], 相比，我們的調查不僅包括文獻中的額外最新論文，而且還批判性地審視了現有技術，突顯了其限制和採用的潛在挑戰。此外，我們討論了新的觀察結果和對抗性示例在數據流形 [90] 中的佈置與對抗性攻擊的韌性的相關性。

2.4 比較總結

總結來說，與相關研究相比，我們的調查以其全面且系統的方法而聞名。我們提出了一種完整的ML-NIDS分類法、針對它們的對抗性攻擊以及潛在的防禦措施。這項分析不僅僅是列舉方法；我們評估了它們的優點和缺點，並提供總結以指導實踐者。我們還探討了這些攻擊及其防禦在NIDS環境中的實際應用性。值得注意的是，我們的調查揭示了對抗性範例的存在和放置以及NIDS中惡意軟體特徵之間相互依存的新的洞見。透過分析這些方面並進行全面審查，我們提取了寶貴的經驗教訓，並提出了一套開發穩健ML-NIDS的指導原則。表1總結了我們的調查與最相關相關研究的比較，並突顯了現有調查中對抗性範例針對ML-NIDS的顯著差距。調查在鞏固知識和確定研究方向方面發揮著不可或缺的作用，這支持了學科的發展。我們相信我們的工作提供了詳細的分析和新穎的洞見，使其成為新進和經驗豐富的NIDS實踐者都寶貴的資源。

3 背景

本節提供關於入侵偵測系統、深度神經網路、學習系統的常見對抗性攻擊以及現有對策的背景資訊。

3.1 入侵偵測系統

入侵偵測系統包括網路流量監控（NIDS）和系統行為分析（主機基礎IDS）以偵測網路攻擊 [6]。它們與其他安全機制（包括防火牆）協同工作，並使用分為兩種主要類型的各種偵測技術：

- 基於簽名的偵測使用從先前攻擊收集的信息和已知漏洞來創建一個具有特徵且可識別的簽名。由於攻擊者不斷更新他們的攻擊模式和方法，簽名很快過時並需要定期更新以保持有效 [71]。由於依賴於已知的攻擊模式，基於簽名的偵測通常提供高精確度與低偽陽性率，但無法偵測新的、未知的攻擊。
- 基於異常的偵測模擬系統和網路的典型行為，並偵測與此行為偏差的潛在攻擊。這種方法的主要優勢在於它能在只要導致可偵測行為變化的情況下識別已知和未知攻擊。此外，一旦計算了系統/網路的典型行為，基於異常的IDS就不需要持續更新。然而，與基於簽名的方法相比，基於異常的偵測通常會產生更高的偽陽性率，使其處理偽報警更耗時。

3.2 智能入侵偵測系統

網路攻擊的日益複雜化促使入侵偵測系統（IDS）取得顯著進步。近年來，整合機器學習技術於IDS的趨勢日益強勁，大幅提升了偵測能力 [127]。

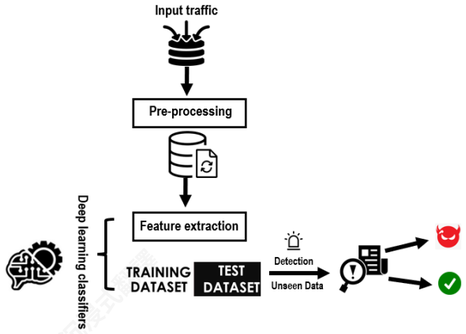


Fig. 2. High-level architecture of a deep learning based-IDS

A growing body of research highlights that ML consistently outperform traditional IDS methods in various settings [6], with DL algorithms, in particular, showing promising results. This advantage arises from the ability of DL models to automatically learn and select descriptive features, eliminating the dependency on complex, expert-driven feature engineering (see Figure 2) — a major limitation in traditional ML [33]. Additionally, DL models excel at handling high-dimensional data and identifying intricate relationships between features, resulting in improved classification accuracy [93]. They also demonstrate strong adaptability to the dynamic nature of network traffic and evolving attack patterns. While DL training can be computationally expensive, advancements in software and hardware are mitigating this challenge [49]. Despite their numerous advantages, however, both ML- and DL-NIDS share a significant vulnerability: adversarial attacks [86, 101]. Adversarial attacks involve adversarial perturbation of training or query data designed to cause the model to fail in detecting malicious activity, thereby undermining its effectiveness.

3.3 Deep Neural Networks

Deep neural networks (DNNs) are algorithms developed to autonomously recognize, extract, and learn key patterns and relationships from input data, eliminating the need for extensive feature engineering by domain experts. DNNs excel at processing large volumes of high-dimensional data, and have consistently redefined state-of-the-art performance across various learning tasks. Practically, a DNN can be represented as a function (or a set of functions) that given an input provides an output as follows:

$$f \in F : \mathbb{R}^n \rightarrow \mathbb{R}^m$$

Deep learning system comprises of two stages, the training and inference stages.

- **The training stage:** Let X be the class of inputs and Y a label space. The learning procedure for a DNN model f consists of the selection of a set of parameters θ that minimize a loss function l on the training set $\mathcal{D} = (\mathcal{X}, \mathcal{Y})$ which measures how well the DNN has learned over the data. The learning the parameters θ occurs through an iterative process where the loss function l, which measures the performance of the learned function f, provides the necessary information to optimize the parameters until an optimum is reached. More formally, neural network training seeks to optimize the following problem:

$$\underset{\theta}{argmin} \sum_{(x,y)\sim\mathcal{D}} l(\theta, x, y)$$

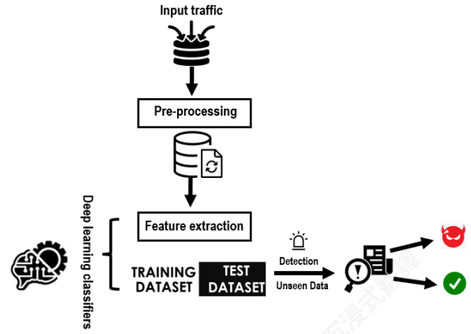


圖2. 基於深度學習的IDS高階架構

越來越多的研究指出，機器學習在多種場景下持續超越傳統IDS方法，特別是深度學習算法，展現出令人期待的成果。這種優勢源於深度學習模型能自動學習並選擇描述性特徵，免除了對複雜、由專家驅動的特徵工程的依賴（見圖2）——這是傳統機器學習的主要限制 [33]。此外，深度學習模型擅長處理高維數據，並識別特徵間複雜的關係，從而提升分類準確性 [93]。它們也展現出對網路流量動態性及演變攻擊模式的強大適應能力。雖然深度學習訓練可能計算成本高昂，但軟體和硬體的進步正逐漸緩解這個挑戰 [49]。然而，儘管機器學習和深度學習基礎的IDS各有優勢，但它們都共享一項顯著的弱點：對抗性攻擊 [86, 101]。對抗性攻擊涉及對訓練或查詢數據進行對抗性干擾，旨在導致模型無法偵測惡意活動，從而削弱其有效性。

3.3 深度神經網路

深度神經網路（DNNs）是為了能自主辨識、萃取及學習輸入數據中的關鍵模式與關係而開發的演算法，無需領域專家進行大量的特徵工程。DNNs擅長處理龐大的高維度數據，並在各種學習任務中持續重新定義先進性能。實際上，DNN可以表示為一個函數（或一組函數），其給定輸入後提供輸出，如下所示：

$$f \in F : \mathbb{R}^n \rightarrow \mathbb{R}^m$$

深度學習系統包含兩個階段，即訓練階段和推論階段。

- 訓練階段：讓 X 為輸入類別， Y 為標籤空間。DNN模型 f 的學習程序包含選擇一組參數 θ，使其在訓練集 $\mathcal{D} = (\mathcal{X}, \mathcal{Y})$ 上最小化損失函數 l，訓練集衡量DNN在數據上學習的成效。參數 θ 的學習透過迭代過程發生，其中損失函數 l（衡量學習函數 f 的性能）提供必要的資訊來優化參數，直到達到最佳點。更形式化地說，神經網路訓練旨在優化以下問題：

$$\underset{\theta}{argmin} \sum_{(x,y)\sim\mathcal{D}} l(\theta, x, y)$$

where θ are the parameters of the DNN to optimize, x and y are the training datapoints and their corresponding labels sampled from the training data $\mathcal{D}$, and l is the loss function used for training.

- **The inference stage:** After finding an optimal set of parameters θ, the model can be used to classify new data. An unseen input x_j and its corresponding label y_j are calculated as follows:

$$y_j = f(\theta, x_j)$$

Additionally, during the classification task, the output of the last layer Y (called *logits*) are transformed into a probability distribution by an activation function (e.g., Softmax):

$$0 \leq Y_i \leq 1 \wedge Y_1 + ... + Y_m = 1;$$

where Y_i can be understood as the probability that input x has class i. The label that has the highest probability $argmax(Y_i)$ presents the class of the input x.

3.4 Adversarial Threats

Traditional ML and DL techniques have been show to be vulnerable to adversarial attacks: manipulations of input data that lead models to produce wrong predictions. One particularly concerning form of such attacks are *adversarial examples* [55, 134]. Initially studied in image classification task, adversarial examples involve subtle, often imperceptible perturbations to the input data that can significantly alter DNN's output. Generally, the training process of a DNN involves minimizing a loss function l, which quantifies the error between the target and the predicted classes. This optimization process aims to reduce the error rate, improve the model's accuracy by adjusting its parameters based on the computed loss. The overall error rate used to evaluate the model's performance is then calculated by comparing the sum of the loss between the target and the corresponding labels.

$$\sum_{(x,y)\sim\mathcal{D}} l(f(\theta, x), y)$$

Adversarial examples are crafted by adding a small *adversarial perturbation* to a clean input x. The details on how the perturbation is generated depend on the specific algorithm $\mathcal{A}$ used, but for DNNs, they typically rely on gradient computation to increase the loss (i.e., error) of the model on a specific input x:

$$x_{adv} = (x + \mathcal{A}(\theta, x, y))$$

Over the years, numerous adversarial attacks have been introduced, and they can be categorized based on four main criteria:

(1) **Attack Timing:** attacks are categorized according to when they happen. Poisoning attacks exploit the model's vulnerability during the training phase by injecting manipulated data in the training set, aiming to corrupt the learning process [140]. Evasion attacks, such as adversarial examples, occur during the test phase, where adversarial perturbations are introduced into the query sample to deceive the model into making incorrect predictions [50, 95, 150].

(2) **Adversarial Goal:** adversarial attacks can be either targeted or untargeted. Targeted attacks aim to misclassify the input as a specific incorrect class. Untargeted attacks aim to force the model to misclassify the input, without specifying which class it should be classified as.

(3) **Adversary's Knowledge:** adversary capabilities are classified based on the level of access to information about the target model.

- *White box-attacks:* the attacker has full knowledge about the model (architecture, training data, model parameters) and can exploit this information to craft highly effective adversarial examples. Many white-box adversarial attacks generation methods exist, such as Limited-memory Broyden–Fletcher–Goldfarb Shanno [134], Fast Gradient Sign Method [55], Basic Iterative Method, Projected Gradient Descent [82], Deepfool [106], Carlini & Wagner [28], and Jacobian-based Saliency Map Attack [87].
- *Black-box attacks:* the attacker has no knowledge of the model's architecture, parameters, or training data, and can only interact with it in an oracle-like fashion by querying specific inputs and observing

其中 θ 為 DNN 需要優化的參數， x 和 y 為從訓練數據 D 中採樣的訓練數據點及其對應標籤，而 l 則是訓練所使用的損失函數。

- 推論階段：在找到一組最佳參數 θ 之後，模型即可用於分類新數據。一個未見過的輸入 x_j 及其對應的標籤 y_j 計算方式如下：

$$y_j = f(\theta, x_j)$$

此外，在分類任務中，最後一層 Y （稱為logits）會透過激勵函數（例如Softmax）轉換成為一個機率分佈：

$$0 \leq Y_i \leq 1 \wedge Y_1 + ... + Y_m = 1;$$

在哪裡 Y_i 可以被理解為輸入 x 有類別 i 的機率。擁有最高機率 $argmax(Y_i)$ 的標籤呈現了輸入 x 的類別。

3.4 對抗性威脅

傳統的機器學習與深度學習技術已被證明容易受到對抗性攻擊：對輸入數據的操縱，導致模型產生錯誤的預測。其中一種特別令人擔憂的攻擊形式是對抗性範例 [55, 134]。最初在圖像分類任務中研究，對抗性範例涉及對輸入數據進行微妙、通常難以察覺的變動，這些變動可以顯著改變深度神經網絡的輸出。通常，深度神經網絡的訓練過程涉及最小化一個損失函數 l，該函數量化了目標類別與預測類別之間的誤差。這個優化過程旨在通過根據計算出的損失來調整其參數，以降低誤差率並提高模型的準確性。然後，通過比較目標與對應標籤之間損失的總和，來計算用於評估模型性能的整體誤差率。

$$\sum_{(x,y)\sim\mathcal{D}} l(f(\theta, x), y)$$

對抗樣本係透過向乾淨輸入 x添加一個小的對抗干擾來製造的。關於如何產生干擾的細節取決於所使用的特定算法A，但對於深度神經網路（DNN），它們通常依賴梯度計算來增加模型在特定輸入 x上的損失（即，錯誤）：

$$x_{adv} = (x + \mathcal{A}(\theta, x, y))$$

多年來，已推出許多對抗性攻擊，並可根據四個主要標準進行分類：

(1) 攻擊時機：攻擊根據其發生時間進行分類。中毒攻擊利用模型在訓練階段的弱點，透過向訓練集注入被操縱的數據來損害學習過程 [140]。迴避攻擊，如對抗樣本，則發生在測試階段，在查詢樣本中引入對抗干擾以騙取模型做出錯誤預測 [50, 95, 150]。

(2) 對抗目標：對抗攻擊可以是針對性的或非針對性的。針對性攻擊旨在將輸入錯誤分類為特定的錯誤類別。非針對性攻擊旨在強迫模型錯誤分類輸入，而無需指定其應該被分類為哪個類別。

(3) 敵方知識：敵方能力根據對目標模型的資訊存取層級進行分類。

- 白盒攻擊：攻擊者對模型有完全的了解（架構、訓練資料、模型參數），並可利用這些資訊製造高度有效的對抗樣本。許多白盒對抗攻擊生成方法存在，例如 Limited-memory Broyden–Fletcher–Goldfarb Shanno [134], Fast Gradient Sign Method [55], Basic Iterative Method,Projected Gradient Descent [82],Deepfool [106], Carlini & Wagner [28],以及 Jacobian-based Saliency Map Attack [87]。
- 黑盒攻擊：攻擊者對模型的架構、參數或訓練資料毫無知識，只能透過詢問特定輸入並觀察來以類神谕的方式與其互動。

the corresponding outputs. Consequently, black-box attack typically rely on iterative methods, where the adversarial sample is updated over time to achieve the desired adversarial goal. These attacks often employ techniques like Generative Adversarial Networks (GAN) [54, 95] or Zeroth-Order Optimization attacks [31], but can also utilize on surrogate models by approximating the target model's behavior and then use these learned models to generate adversarial inputs for the original system [157].

- *Grey-box attacks:* the attacker has partial knowledge about the DNN model. Unlike black-box access, where only the inputs and outputs of the model are observable, grey-box access allows some limited understanding or control over the model's internal structure or behavior. This can include access to certain layers or parameters but not full transparency like white box access, where the entire model's structure and training process are fully known and accessible.

(4) **Attack Frequency:** adversarial attacks can be one-shot attacks or iterative attacks. One-shot attacks generate adversarial perturbations are through a single-step algorithm. Iterative adversarial attacks rely on multiple iterations, repeatedly querying the classifier to refine the adversarial perturbation.

3.5 Adversarial Countermeasures

Given the relevance of adversarial examples and the threat they pose to the integrity of ML and DL applications, several defense mechanisms have been explored over the years. Several strategies have been explored, such as adversarial training, model regularization, input preprocessing, and robust optimization, aiming to enhance the resilience and reliability of neural networks against such threats. Despite much research effort, however, most proposals have proven ineffective and, over time, have succumbed to novel adaptive attacks. Figure 3 presents the timeline of these defenses and corresponding adaptive attacks that reveal their limitations [15, 25–27, 142]. Adversarial example countermeasures can be divided into two main categories:

(1) **Adversarial Training:** the goal of adversarial training approaches is to enable the model to develop a robust representation of the training data, thereby reducing the impact of adversarial examples. This is accomplished by training the model using both clean and adversarial datapoints, allowing for an adaptation of the learned decision boundary to accurately classify both perturbed and unperturbed instances [55]. Formalizing, adversarial training aims to find a set of model parameters θ that minimize the following expression:

$$\underset{\theta}{argmin}\ E_{(x,y)\sim D}[\underset{\delta\in\Delta}{max}\ l(\theta, x+\delta, y)] \quad (4)$$

where δ is an adversarial perturbation, Δ represents the set of all allowable adversarial perturbations, D denotes the data distribution over inputs $x \in \mathbb{R}^d$ and labels y, $l(\theta, x, y)$ is the loss function for the model parameters θ.

(2) **Architecture Reinforcing:** this category of countermeasures includes several methods that share the common purpose of reinforcing the model's architecture, each employing different strategies as detailed below.

- ***Detection and Rejection:*** this method involves using trained detectors to evaluate model inputs and detect potential adversarial perturbations. When a perturbation is detected, the system executes a series of specialized operations to reduce/eliminate the identified threat [102].
- ***Input Transformation:*** this approach applies a set of transformations to the input data to eliminate potential adversarial perturbations before the sample is used to query the model. The input transformations typically aim at disrupting the features altered by adversarial perturbations, mitigating the effect of the attacks [8, 19].

對應的輸出。因此，黑盒攻擊通常依賴迭代方法，其中對抗樣本會隨著時間更新以達成預期的對抗目標。這些攻擊常使用生成對抗網絡（GAN） [54, 95] 或零階優化攻擊 [31], 等技術，但也可以透過近似目標模型的行為來利用替代模型，然後使用這些學習到的模型為原始系統生成對抗輸入 [157]。

- 灰盒攻擊：攻擊者對DNN模型有部分知識。與黑盒訪問不同，黑盒訪問僅能觀察模型的輸入和輸出，而灰盒訪問允許對模型內部結構或行為進行有限的了解或控制。這可能包括訪問某些層或參數，但無法像白盒訪問那樣完全透明，白盒訪問中模型的整個結構和訓練過程都是完全已知且可訪問的。

(4) 攻擊頻率：對抗攻擊可以是單次攻擊或迭代攻擊。單次攻擊通過單步算法生成對抗擾動。迭代對抗攻擊依賴多次迭代，反覆查詢分類器以精煉對抗擾動。

3.5 對抗應對措施

考慮到對抗範例的相關性以及它們對機器學習和深度學習應用程式完整性的威脅，多年來已探索了多種防禦機制。已探索的策略包括對抗訓練、模型正則化、輸入預處理和穩健優化，旨在增強神經網路對此類威脅的韌性和可靠性。然而，儘管投入了大量研究努力，但大多數提議已被證明無效，並且隨著時間的推移，已經屈服於新的自適應攻擊。圖3展示了這些防禦措施及其相應的自適應攻擊的時間線，揭示了它們的局限性 [15, 25–27, 142]。對抗範例應對措施可分為兩個主要類別：

(1) 對抗訓練：對抗訓練方法的目的在於讓模型能夠發展出對訓練數據的穩健表示，從而減少對抗性範例的影響。這是通過使用乾淨與對抗性數據點來訓練模型來實現的，允許學習到的決策邊界進行適應，以準確分類受干擾與未受干擾的實例 [55]。形式化地說，對抗訓練旨在找到一組模型參數 θ ，以最小化以下表達式：

$$\underset{\theta}{argmin}\ E_{(x,y)\sim D}[\underset{\delta\in\Delta}{max}\ l(\theta, x+\delta, y)] \quad (4)$$

where δ 是對抗性干擾， Δ 代表所有允許的對抗性干擾的集合， D表示輸入 $x \in \mathbb{R}^d$和標籤 y上的數據分佈， $l(\theta, x, y)$是模型參數 θ的損失函數。

(2) 架構加強：這類反制措施包含多種方法，它們的共同目的是加強模型的架構，每種方法採用不同的策略，詳情如下。

- 檢測與拒絕：這種方法涉及使用訓練好的檢測器來評估模型輸入並檢測潛在的對抗性干擾。當檢測到干擾時，系統執行一系列專門的操作來減少/消除識別出的威脅 [102]。
- 輸入轉換：這種方法對輸入數據應用一組轉換，以在樣本用於查詢模型之前消除潛在的對抗性干擾。輸入轉換通常旨在破壞對抗性干擾所改變的特徵，減輕攻擊的影響 [8, 19]。

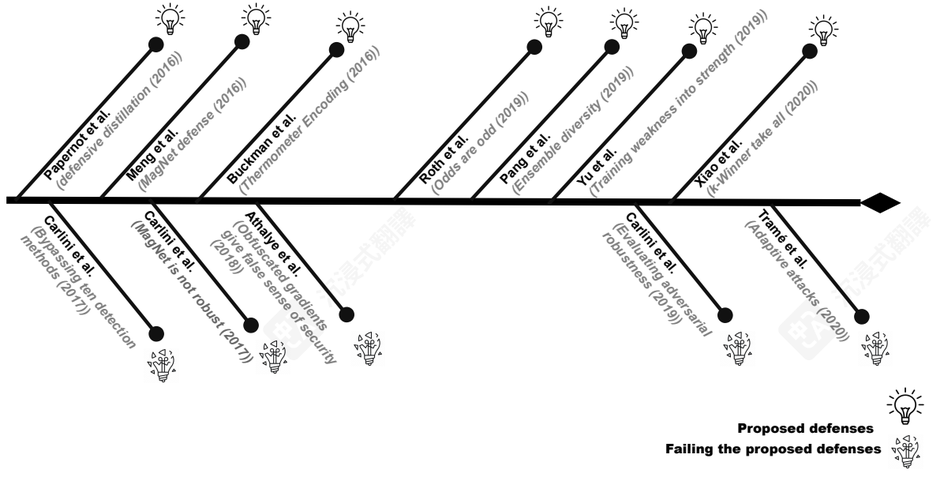


Fig. 3. Timeline of adversarial attacks and defenses.

- ***Gradient Masking:*** this technique relies on hiding the internal state of the model — its gradient on specific inputs. Since most adversarial example algorithms require gradient computation, the attacks are ineffective if gradient information cannot be obtained accurately [32].
- ***Manifold Projection:*** this approach leverages the natural data distribution (manifold) to detect and remove adversarial perturbations [90, 132]. Adversarial examples typically reside in different region of the domain compared to clean samples, and manifold projection can employed to discard adversarial examples that deviate significantly from the natural data structure.
- ***Ensemble Method:*** this approach relies on the combination of multiple models and model diversity to make it more challenging for attackers to generate effective adversarial examples [114, 141].

4 LITERATURE REVIEW OF INTELLIGENT NETWORK INTRUSION DETECTION SYSTEMS

Machine learning, especially deep learning, shows promise in NIDS. While traditional methods like Support Vector Machines (SVM), Decision Trees (DTs), and Naive Bayes (NB) are simpler and easier to interpret, they perform poorly against complex attacks. DL-NIDS are more effective but require more resources and are less transparent in their decision-making. This section presents a detailed review of both traditional and DL-NIDS, categorized by learning approach (single vs. multimodal). This is summarized comprehensively in Table 2.

4.1 Single Learning

Single Learning refers to a machine learning approach where a single, monolithic model is used to perform anomaly detection or classification on unseen data.

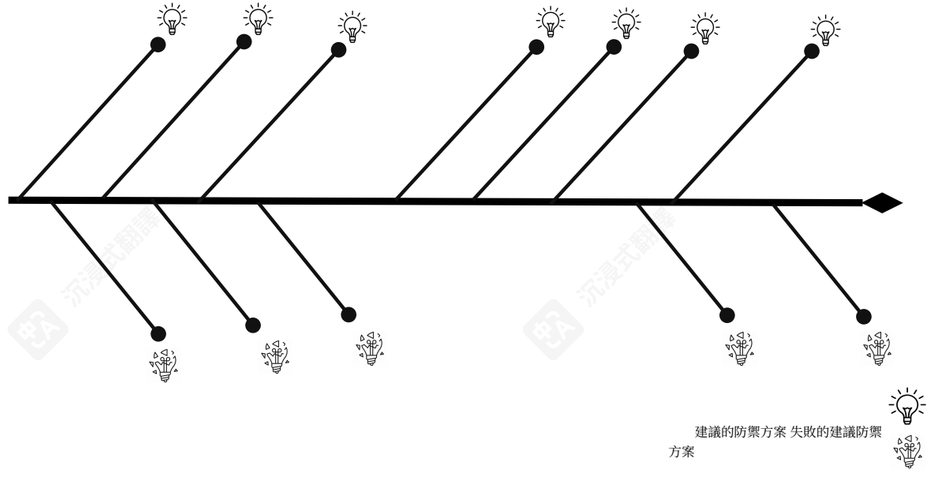


圖 3。對抗攻擊與防禦的時間線。

- 梯度遮罩：此技術依賴於隱藏模型的內部狀態——其在特定輸入上的梯度。由於大多數對抗示例算法需要梯度計算，因此如果無法準確獲取梯度信息，攻擊就無效 [32]。
- 流形投影：此方法利用自然數據分佈（流形）來檢測和移除對抗干擾 [90, 132]。對抗示例通常位於與乾淨樣本不同的域區域，而流形投影可以應用於拋棄顯著偏離自然數據結構的對抗示例。
  與乾淨樣本相比，流形投影可以應用於拋棄顯著偏離自然數據結構的對抗示例。
- 集合方法：此方法依賴於多個模型的組合與模型多樣性，讓攻擊者更難生成有效的對抗樣本 [114, 141]。

4 文獻回顧：智能網路入侵偵測系統

機器學習，尤其是深度學習，在NIDS（網絡入侵檢測系統）方面顯示出前景。雖然傳統方法如支持向量機（SVM）、決策樹（DTs）和朴素貝葉斯（NB）更簡單且容易解釋，但它們在對抗複雜攻擊時表現不佳。DL-NIDS（基於深度學習的NIDS）更有效，但需要更多資源，且在決策制定方面透明度較低。本節詳細介紹了傳統NIDS和DL-NIDS，按學習方法（單一對多模態）進行分類。這一點在表2中全面總結。

4.1 單一學習

單一學習是指一種機器學習方法，使用單一、整體的模型來對未見數據進行異常檢測或分類。

4.1.1 Traditional Machine Learning Approaches: Research persists on traditional ML methods for NIDS due to their interpretability compared to DL, aiming to bridge the performance gap with DNN. Chkirbene et al. [34] developed two NIDS models based on a hybrid feature selection approach: Trust-based Intrusion Detection and Classification System (TIDCS), and an accelerated variant (TIDCS-A). Their method combines randomization, wrapper, and filter strategies to identify the most descriptive features to use for classification. First, TIDCS randomly selects multiple subsets of features and trains a Random Forest (RF) classifier on each subset. Then, the trained models are used to compute the score of individual features in each subset based on the accuracy achieved by the classifier. Finally, the best-scoring features across all subsets are selected for training the final model. Despite achieving good classification results, the reliance on pre-trained model accuracy for feature selection can introduce bias in the classification. Moreover, focusing only on increasing accuracy during feature selection can lead to decreased robustness, making the model more susceptible to evasion attacks.

Alahmadi et al. [4] propose a bot identification method that combines Markov Chains and RF classifiers. By modeling network behavior as state transitions in the Markov Chain, they capture high-level protocol states as features for botnet family detection. This focus on high-level behavior enhances robustness, as modifying the malware logic would be necessary for evasion. However, similar techniques based on system call transition analysis have been demonstrated to remain vulnerable to evasion attacks [109]. Moreover, potential overfitting on specific network patterns and the lack of cross-dataset validation require further investigation. For detecting Denial of Service (DoS) attacks in Software-Defined Networks (SDN), Ramamurthy et al. [99] suggest an in-controller NIDS that leverages control packets generated by SDN when encountering previously unseen network flows. The proposed NIDS analyzes packets' data distribution, such as IP addresses, ports, and timing, using an entropy metric for anomaly detection with an SVM model. Although this transparent method simplifies SDN integration, its reliance on easily manipulated features makes it susceptible to evasion attacks.

TAKEAWAY. Traditional ML in NIDS often requires complex and sophisticated feature selection and dimensionality reduction for optimal performance. Although this increases the complexity of the NIDS architecture, it enables a more focused classification on specific aspects of the network traffic. However, complex designs may result in biases and poor generalization to unseen data. Moreover, although selected features are often descriptive, they generally lack resilience to adversarial attacks due to the challenges in creating robust selection and reduction strategies.

4.1.2 Deep Learning Approaches: To address limitations of traditional ML, recent NIDS research prioritizes DL. This focus employs DNN's ability to learn complex features from data, aiming for better classification. This focus can be divided into two categories: fully unsupervised feature representation and guided feature representation.

Fully Unsupervised Feature Representation. This family of works uses end-to-end DNN to autonomously learn feature representations from unlabeled data and perform classification. In [160], the authors developed DFEL, a NIDS based on a DL approach for feature embedding learning. They pre-trained a DNN on historical NIDS data (normal and abnormal traffic), using it as a feature extractor for IoT traffic. These extracted features were then used to train another DNN for anomaly detection, leveraging the generalist DNN's data representation to classify IoT network traffic.

Similarly, Manimurugan et al. [92] use a Deep Belief Network (DBN) to learn descriptive features from raw network traffic. They combine unsupervised pre-training with supervised fine-tuning to detect IoT network attacks. The DBN learns effective feature representations during the unsupervised phase, which are then used for classification in the fine-tuned model. In line with [92, 160], Alom et al. [9] developed a DNN-NIDS based on neuromorphic computing systems by combining unsupervised and supervised training to learn descriptive feature representations and perform classification. An AE is used for unsupervised feature learning from unlabeled data, which is then fed to a fully-connected network trained on labeled data. The trained DNN's weights are quantized and mapped to an IBM TrueNorth neuromorphic SoC for evaluation.

4.1.1 傳統機器學習方法：由於傳統ML方法相較於DL具有可解釋性，因此對NIDS的研究持續進行，旨在縮小其與DNN的性能差距。Chkirbene等人 [34]基於混合特徵選擇方法開發了兩個NIDS模型：基於信任的入侵檢測和分類系統（TIDCS），以及一個加速變體（TIDCS-A）。他們的方法結合了隨機化、包裝器和過濾策略來識別最描述性的特徵以用於分類。首先，TIDCS隨機選擇多個特徵子集，並在每個子集上訓練一個隨機森林（RF）分類器。然後，使用訓練好的模型根據分類器達到的準確率計算每個子集中個別特徵的得分。最後，選擇所有子集中得分最高的特徵來訓練最終模型。儘管取得了良好的分類結果，但依賴預訓練模型準確率進行特徵選擇可能會導入分類中的偏差。此外，在特徵選擇期間僅關注提高準確率可能會導致穩健性下降，使模型更容易受到規避攻擊。

Alahmadi 等人 [4]提出一種結合馬爾可夫鏈和RF分類器的機器人識別方法。透過將網路行為建模為馬爾可夫鏈中的狀態轉換，他們將高層級協議狀態作為特徵，用於識別機器人家族。這種對高層級行為的關注增強了其穩健性，因為為了迴避攻擊，必須修改惡意軟體邏輯。然而，基於系統調用轉換分析的類似技術已被證明仍然容易受到迴避攻擊 [109]。此外，對特定網路模式的潛在過度擬合以及缺乏跨數據集驗證需要進一步研究。為了在軟體定義網路（SDN）中偵測拒絕服務（DoS）攻擊，Ramamurthy 等人 [99] 建議一種控制器內NIDS，該NIDS利用SDN在遇到先前未見過的網路流量時生成的控制封包。所提出的NIDS使用熵度量標準，透過SVM模型分析封包的數據分佈，例如IP位址、埠和時間，進行異常偵測。儘管這種透明的方法簡化了SDN整合，但其對易於操縱的特徵的依賴使其容易受到迴避攻擊。

TAKEAWAY. 傳統的NIDS常需要複雜且精密的特徵選擇與降維，以達到最佳性能。雖然這增加了NIDS架構的複雜度，但能夠針對網路流量特定面向進行更專注的分類。然而，複雜的設計可能導致偏差，並且對未見過的數據泛化能力不佳。此外，雖然選擇的特徵通常具有描述性，但因為建立穩健選擇與降維策略的挑戰，它們通常缺乏對抗性攻擊的韌性。

4.1.2 深度學習方法：為了彌補傳統機器學習的局限性，近期的NIDS研究優先考慮深度學習。這種重點利用DNN從數據中學習複雜特徵的能力，旨在進行更好的分類。這種重點可分為兩個類別：完全無監督的特徵表示與指導式特徵表示。

完全無監督的特徵表示。這類研究使用端到端DNN從無標籤數據中自主學習特徵表示，並進行分類。在 [160],中，作者們開發了DFEL，一個基於深度學習方法的特徵嵌入學習NIDS。他們在歷史NIDS數據（正常和異常流量）上預訓練一個DNN，將其作為IOT流量的特徵提取器。提取的特徵之後用於訓練另一個DNN進行異常檢測，利用這個通用DNN的數據表示來分類IOT網路流量。

類似地，Manimurugan 等人 [92] 使用深度信念網絡 (DBN) 來從原始網路流量中學習描述性特徵。他們結合無監督預訓練與監督微調來偵測 IoT 網路攻擊。DBN 在無監督階段學習有效的特徵表示，這些特徵之後用於微調模型的分類。與此同時， [92, 160], Alom 等人 [9] 基於神經計算系統開發了一個 DNN-NIDS，透過結合無監督和監督訓練來學習描述性特徵表示並進行分類。一個自組合網絡用於從未標記數據中進行無監督特徵學習，然後輸入到在標記數據上訓練的全連接網絡。訓練的 DNN 的權重被量化並映射到 IBM TrueNorth 神經計算 SoC 上進行評估。

Fully unsupervised feature representation approaches enable the utilization of available unlabeled data — such as general network traffic — to learn useful features for similar domains with limited labeled data (e.g., IoT), ultimately improving the performance of the final classifier. However, relying on fully unsupervised feature learning risks overfitting on characteristics that are specific to the source domain but absent in the target domain, introducing biases in the final model. Furthermore, exclusively relying on unsupervised feature learning increases vulnerability to adversarial attacks, as unconstrained (or semi-constrained) features can be more easily exploited for evasion [117].

Guided Feature Representation. This family of works uses a combination of DNN and other techniques to learn descriptive traffic embeddings for intrusion detection. Laghrissi et al. [83] combine Long Short-Term Memory (LSTM) with Principal Component Analysis (PCA) and Mutual Information (MI) for dimensionality reduction and feature selection. The approach first uses PCA to transform correlated features into a smaller set of uncorrelated features. Then, MI is employed to compute the statistical dependence between individual features and their labels, selecting those highly relevant to the target class. The model is then trained on these features for both binary and multiclass classification. In [79], Kan et al. developed a CNN-NIDS using adaptive particle swarm optimization (APSO) to tune the model's hyperparameters. They use a custom network dataset from various IoT devices and manually select aggregate statistical features of the traffic (e.g., correlation coefficients on packet sizes, mean, variance, etc.). The feature vector is then fed to a 1-dimensional CNN for feature learning and classification. APSO is used to optimize the CNN's hyperparameters based on a held-out validation set to evaluate the fitness of the offsprings. Thakkar et al. [138] use a fusion-based method on statistical features derived from raw traffic data. Features for training are selected based on a ranking system considering standard deviation, mean, and median. They are added iteratively in ranking order until the model's accuracy no longer improves. Piskozub et al. propose MalPhase in [121], a multi-phase technique for anomaly detection, malware type classification, and malware family classification using augmented NetFlow data. It combines expert knowledge for feature selection with AE-based feature representation learning to extract descriptive features. The AE is trained to reconstruct original malicious data by combining benign and malicious traffic, forcing it to learn core features of malware traffic. MalPhase uses a multi-tiered architecture: lower-tier models analyze short time windows for brief spikes in activity, while higher-tier models identify sustained malware communication. In [156], Zhong et al. combine had-crafted feature engineering, unsupervised feature learning, and supervised DNN-based classification to identify malicious network traffic. Damped Incremental Statistics [104] is used to extract features from flow entries of SDN switches, and a fully connected AE is used to further refine the feature representation. The trained autoencoder is used to obtain an "anomaly score" and identify potential anomalous traffic. The anomalous traffic is then used to train an LSTM network and classify unseen network traffic as either normal or malicious.

The main advantage of combining DL with statistical and manual feature selection techniques lies in the integration of DNN's ability to identify non-linear correlations in data with domain-expert knowledge. This combination aids in identifying the most descriptive features, while leveraging prior expert knowledge to mitigate biases in the training data. Furthermore, combining expert knowledge with DNN for feature representation can enhance robustness against evasion attacks, provided that the features are well-designed to limit manipulation. However, with a partial exception of [121], none of the previously mentioned works consider robustness in their feature representation and selection designs. Additionally, their threat models do not account for sophisticated evasion strategies, such as adversarial examples or adaptive attacks.

TAKAWAY. DL-NIDS show several advantages over traditional ML-NIDS. DNNs can reduce the need for human intervention by learning unsupervised feature representations. They can also identify complex, non-linear relations in the training data, which typically results in improved performance compared to traditional ML techniques. Despite adversarial noise, unsupervised DNN techniques can be leveraged to learn robust embeddings from unlabeled data [121], which improves robustness. However, DL-NIDS are susceptible to adversarial attacks

完全無監督的特徵表示方法能夠利用可用的未標記數據——例如一般網路流量——來為類似領域學習有用的特徵，當標記數據有限時（例如 IoT），最終提高最終分類器的性能。然而，依賴完全無監督的特徵學習會導致過度擬合來源領域的特定特性，但在目標領域中不存在，這會在最終模型中引入偏差。此外，專注於無監督特徵學習會增加對對抗性攻擊的脆弱性，因為不受約束（或半受約束）的特徵更容易被利用來逃逸 [117]。

導引特徵表示。這個系列的著作使用 DNN 和其他技術的組合來學習用於入侵偵測的描述性交通嵌入。Laghrissi 等人 [83] 將長短期記憶 (LSTM) 與主成分分析 (PCA) 和互信息 (MI) 結合起來進行降維和特徵選擇。這種方法首先使用 PCA 將相關特徵轉換為一組較小的無相關特徵。然後，使用 MI 計算個別特徵與其標籤之間的統計相依性，選擇那些與目標類別高度相關的特徵。然後在這些特徵上訓練模型進行二元和多類別分類。在 [79], 中，Kan 等人開發了一個使用自适应粒子群优化 (APSO) 來調整模型超參數的 CNN-NIDS。他們使用來自各種 IoT 裝置的自定義網絡數據集，並手動選擇流量（例如，包大小的相關係數、平均值、變異數等）的聚合統計特徵。然後將特徵向量輸入到一維 CNN 中進行特徵學習和分類。使用 APSO 根據保留的驗證集來優化 CNN 的超參數，以評估後代的適合度。
Thakkar 等人 [138] 對從原始流量數據中衍生的統計特徵使用基於融合的方法。選擇用於訓練的特徵基於一個考慮標準差、平均值和中位數的排名系統。他們按排名順序迭代地添加，直到模型的準確性不再提高。Piskozub 等人在 [121], 中提出 MalPhase，這是一種多階段技術，用於異常偵測、惡意軟件類型分類和惡意軟件家族分類，使用增強的 NetFlow 數據。它結合專家知識進行特徵選擇，以及基於 AE 的特徵表示學習來提取描述性特徵。AE 被訓練來通過結合良性流量和惡意流量重建原始惡意數據，迫使其學習惡意流量核心特徵。MalPhase 使用多層次架構：低層次模型分析短時間窗口中的短暫活動尖峰，而高層次模型識別持續的惡意通信。在 [156], 中，Zhong 等人結合了手動製作的特徵工程、無監督特徵學習和監督的 DNN 進行分類，以識別惡意網絡流量。使用 Damped Incremental Statistics [104] 從 SDN 開關的流入口中提取特徵，並使用一個全連接 AE 進一步精煉特徵表示。訓練的 autoencoder 用於獲得一個“異常分數”並識別潛在的異常流量。然後使用異常流量來訓練一個 LSTM 網絡，並將未見過的網絡流量分類為正常或惡意。

結合深度學習（DL）與統計及手動特徵選擇技術的主要優勢在於整合深度神經網路（DNN）辨識數據中非線性相關性的能力與領域專家知識。這種結合有助於識別最具描述性的特徵，同時利用先驗專家知識來減輕訓練數據中的偏差。此外，將專家知識與DNN結合進行特徵表示可以增強對迴避攻擊的韌性，前提是特徵設計得能夠限制操縱。然而，除了 [121], 之外，先前提到的所有工作在特徵表示和選擇設計中均未考慮韌性。此外，它們的威脅模型也未考慮到複雜的迴避策略，例如對抗性示例或自適應攻擊。

TAKAWAY。DL-NIDS 相較於傳統的 ML-NIDS 總體上有數項優勢。DNNs 可以透過學習非監督式特徵表達來減少人為干預的需求。它們也能夠識別訓練數據中的複雜、非線性關係，這通常導致其表現優於傳統的 ML 技術。儘管存在對抗性雜訊，非監督式的 DNN 技術仍可被利用來從未標註數據中學習穩健的嵌入 [121], ，從而提升其穩健性。然而，DL-NIDS 仍然容易受到對抗性攻擊。

that adversaries can exploit to compromise their performance, limiting their reliability. While some approaches are designed to be robust against simpler attacks, such as noise injection, resilience against sophisticated adversarial attacks is generally lacking. Moreover, DNN-NIDS face challenges due to their black-box nature, as they lack formal methods to analyze their decision-making process. While this drawback can be mitigated by incorporating human expert supervision in the feature selection process, it cannot be entirely eliminated.

4.2 Multimodel Learning

In contrast to single learning, multimodel or ensemble learning combines the prediction of multiple models to improve performance and resilience, reducing the error rate if model errors are independent and uncorrelated.

4.2.1 Combination of Multiple Traditional Machine Learning Algorithms: Several works propose combinations of different traditional ML techniques to address the limitations of any individual approach. Tama et al. [135] propose a feature selection and ensemble classification method for intrusion detection. They employ evolutionary algorithms like particle swarm optimization to identify the most descriptive features. These algorithms evaluate feature subsets based on the classification accuracy of a decision tree. Next, a two-stage ensemble classifier is utilized: a rotation forest acts as a filter, selecting feature subsets for training a bagged ensemble of weak learners. Finally, majority voting combines individual predictions for the final classification. Similarly, Zhou et al. [159] suggest a two-stage approach with feature selection followed by ensemble classification. They employ Correlation-Based Feature Selection for initial selection and an evolutionary algorithm to eliminate redundant features. However, unlike Tama et al. [135], they use a decision tree and random forest ensemble for classification. While the adoption of an ensemble of weaker classifiers is typically more resilient to changes in data distribution and tends to generalize better than single-model approaches [46], the overreliance of these approaches on performance-based metrics for feature selection is likely to introduce biases tied to the training data, ultimately compromising generalization capabilities.

In [58], an intrusion detection method combining an ensemble of SVM models with feature augmentation and train set splitting is proposed. The authors use fuzzy-c means clustering to divide training data into overlapping subsets, which are then processed and augmented with multivariate marginal density ratio features. An ensemble of SVM classifiers is trained on the augmented subsets, and their outputs are combined by a separate SVM model for the final classification. Despite achieving good performance on the considered dataset, relying on ensembles of homogeneous models can lead to decreased robustness to adversarial attacks compared to heterogeneous ensembles [114].

TAKAWAY. Ensembles of traditional ML algorithms help improve performance and can sometimes improve generalization to data with different distributions, especially when heterogeneous models are used. However, the use of ensembles leads to increased computational cost and complexity, limiting its applicability to certain settings. Excessive reliance on performance metrics in feature selection can introduce biases, which may hinder the model's ability to generalize effectively to new data. Lastly, while less complex, homogeneous ensembles are generally more vulnerable to adversarial attacks [114].

4.2.2 Combination of Deep Learning and Traditional Machine Learning: Due to the limitations of single-model DL techniques and ML-based approaches, several studies propose hybrid models that combine DL and traditional ML to compensate for each other's weaknesses. Marir et al. [94] proposed a distributed framework for large-scale network intrusion detection that combines a Deep Belief Network (DBN) with a multilayer ensemble SVM. The DBN is utilized as a feature extractor, and the ensemble of SVM is used to classify samples based on majority voting. Although the framework performs well in a variety of network settings, the use of multiple, multi-layer networks is resource-intensive and complex to maintain. Moreover, it is challenging to tune the hyperparameters of the networks without introducing biases tied to the training data. In [74], Imran et al. combine a non-symmetrical

這些漏洞可能被攻擊者利用來損害其性能，限制其可靠性。雖然有些方法設計用來抵禦簡單的攻擊，例如干擾注入，但對於複雜的對手攻擊的韌性通常是缺乏的。此外，DNN-NIDS 由於其黑盒特性面臨挑戰，因為它們缺乏分析其決策過程的正式方法。雖然這個缺點可以通過在特徵選擇過程中結合人類專家監督來緩解，但它不能完全消除。

4.2 複數模型學習

與單一學習相較，複數模型或集成學習結合多個模型的預測以提升性能與韌性，若模型錯誤獨立且無相關性，則可降低錯誤率。

4.2.1 多個傳統機器學習演算法的結合：數項研究提出結合不同傳統機器學習技術，以克服任何單一方法的限制。Tama 等人 [135] 提出一種針對入侵偵測的特徵選擇與集成分類方法。他們運用粒子群優化等進化演算法來識別最具描述性的特徵。這些演算法根據決策樹的分類準確度來評估特徵子集。接下來，利用兩階段集成分類器：旋轉森林作為篩選器，選擇特徵子集用於訓練一組弱學習者的袋狀集成。最後，多數投票結合個別預測以進行最終分類。類似地，Zhou 等人 [159] 建議一種先進行特徵選擇，再進行集成分類的兩階段方法。他們運用基於相關性的特徵選擇進行初始選擇，並使用進化演算法消除重複特徵。然而，與 Tama 等人 [135], 不同的是，他們使用決策樹和隨機森林集成進行分類。雖然採用弱學習者集成的做法通常對數據分佈變化更具韌性，且相較於單一模型方法，通常具有更好的泛化能力 [46], ，但這些方法過度依賴基於性能的指標進行特徵選擇，很可能引入與訓練數據相關的偏差，最終損害泛化能力。

在 [58], 中，提出了一種結合 SVM 模型集合、特徵增強和訓練集分割的入侵偵測方法。作者使用模糊 C 均值聚類將訓練數據分為疊加的子集，然後對這些子集進行處理並使用多變量邊際密度比特徵進行增強。在增強的子集上訓練 SVM 分類器集合，並由一個獨立的 SVM 模型將其輸出結合以進行最終分類。儘管在所考慮的數據集上取得了良好的性能，但與異質集合相比，依賴同質模型的集合可能會導致對對抗性攻擊的穩健性降低 [114]。

TAKAWAY。傳統機器學習算法的集合有助於提高性能，有時甚至可以改善對分佈不同的數據的泛化能力，特別是當使用異質模型時。然而，使用集合會導致計算成本和複雜性增加，限制了其在某些設定中的應用。在特徵選擇中過度依賴性能指標可能會引入偏差，這可能會妨礙模型有效泛化到新數據的能力。最後，雖然複雜性較低，但同質集合通常更容易受到對抗性攻擊 [114]。

4.2.2 深度學習與傳統機器學習的結合：由於單一模型DL技術和基於ML方法的局限性，數項研究提出結合DL和傳統ML的混合模型，以彌補彼此的弱點。Marir等人. [94]提出了一個用於大規模網路入侵偵測的分發式框架，該框架結合了深度信念網絡（DBN）和多層集成SVM。DBN被用作特徵提取器，而SVM的集成則用於基於多數投票對樣本進行分類。儘管該框架在各種網路設定中表現良好，但使用多個、多層網絡資源密集且維護複雜。此外，在不引入與訓練數據相關的偏差的情況下調整網絡的超參數是一項挑戰。在 [74], Imran等人結合了一個非對稱

deep auto-encoder (NDAE) with an SVM for network traffic classification. They use an encoder-only architecture to encode input data into latent representations, omitting reconstruction, which decreases computational time and costs. Similar to other works leveraging AEs [94, 121], the proposed approach demonstrates an improved ability to learn complex feature correlations and identifying significant features. However, the proposed architecture does not account for robustness to adversarial examples and adaptive attacks, and its heavy reliance on unguided, fully unsupervised learning risks overfitting on the specific characteristics of the training data. Ravi et al. [122] propose a recurrent neural network (RNN)-based feature fusion method consisting of two phases: (i) feature learning using an RNN network and dimensionality reduction, and (ii) classification with an heterogeneous ensemble of RF, SVM, and logistic regression model. First, the RNN feature extraction captures temporal correlations in the input data and Kernel Principal Component Analysis reduces the dimensionality of the extracted features. Finally, the classification is carried out by the ensemble through a voting mechanism. While effective, this complex classification pipeline is computationally intensive and challenging to utilize in real-time scenarios. Additionally, features based on temporal correlations are particularly vulnerable to adversarial attacks, making them an easy target for manipulation [1]. Recently, Arreche et al. [14] introduced a two-level ensemble learning framework. The first level of the framework uses base an ensemble of models to generate both a class prediction and a prediction probability. The second level of the framework combines the predictions and the probabilities from the first level models to further refine the classification through a different ensemble of models. The framework employs a variety of model architectures (DNN, DT, KNN, SVM) and boosting techniques for the two ensembles, creating a highly heterogeneous architecture. While the approach shows good performance across several dataset, the framework is computationally intensive, limiting its practicality in real-time scenarios. Furthermore, the complex interplay among the numeours classifiers across the two levels of the framework makes it difficult to explain the decision-making process, and may introduce overfitting and subtle biases tied to the training data.

TAKAWAY. Heterogeneous ensembles combining DL and traditional ML models are generally effective for intrusion detection. DNNs enable the effective use of large (unlabeled) datasets and serve as powerful feature extractors, while traditional ML algorithms provide accurate classification with lower computational requirements. However, relying on fully unsupervised feature learning may lead to increase susceptibility to adversarial attacks, as well as potentially introducing biases tied to the training data. Moreover, the increased architectural complexity of large ensembles introduces restrictions in the applicability of the techniques in resource- or time-constrained scenarios. Finally, the reliance on DL for feature learning and extraction compromises the explainability provided by traditional ML techniques, making it more difficult to interpret the model's decision-making process.

4.2.3 Combination of Multiple Deep Learning Algorithms: To leverage the ability of DNNs to autonomously learn descriptive features and identify non-linear correlations in data, several studies have proposed combining multiple DL models to enhance detection performance. Mirsky et al [104] proposed Kitsune, an AE-based ensemble NIDS. Kitsune is a fully unsupervised approach that leverages Damped Incremental Statistics and AE reconstruction error rate to detect potentially anomalous traffic. The approach computes temporal statistics for each communication channel in the network, which are then mapped and input to a set of autoencoders. The reconstruction error of each AE is further processed by another fully connected AE to produce a final anomaly score for the traffic. While the completely unsupervised nature of Kitsune makes it easier to train and deploy, the reliance on temporal features makes it susceptible to evasion attacks. Additionally, the fully unsupervised nature of the approach further exacerbates this vulnerability. Nandanwar et al. [107] have recently proposed AttackNet, a model based on the combination of CNN and Gated Recurrent Units (GRU) for classifying network time-series data. The proposed architecture utilizes a limited number of gates to mitigate overfitting. Furthermore, the combination of GRU and CNN enables the analysis of both temporal and spatial features of data, improving the detection capabilities of the ensemble. Hasan et Securing industrial internet of things against botnet attacks using hybrid deep learning approachal. [61] tackle intrusion detection within IIoT environments through a hybrid DL strategy combining

最近，Arreche等人. [14]引入了一個兩級集成學習框架。框架的第一級使用基於模型的集成來生成類別預測和預測概率。框架的第二級結合第一級模型的預測和概率，以通過另一個集成模型進一步精煉分類。該框架採用多種模型架構（DNN、DT、KNN、SVM）和提升技術，為兩個集成創建了一個高度異構的架構。雖然該方法在多個數據集上展現出良好的性能，但該框架計算密集，限制了其在實時場景中的實用性。此外，框架中兩級之間的眾多分類器之間的複雜相互作用使得解釋決策過程變得困難，並且可能導致過度擬合和與訓練數據相關的微妙偏差。

TAKAWAY. 組合深度學習與傳統機器學習模型的異質集成體通常對入侵偵測非常有效。DNNs 能夠有效地利用大型（未標註）數據集，並作為強大的特徵提取器，而傳統機器學習算法則提供計算需求較低的準確分類。然而，依賴完全無監督的特徵學習可能會增加對對抗性攻擊的易受性，並且可能引入與訓練數據相關的偏差。此外，大型集成體的架構複雜度增加，導致在資源或時間受限的場景中應用技術時受到限制。最後，依賴深度學習進行特徵學習和提取，損害了傳統機器學習技術所提供的可解釋性，使得解釋模型的決策過程更加困難。

4.2.3 多個深度學習算法的組合：為了利用深度神經網絡（DNN）自動學習描述性特徵和識別數據中的非線性相關性的能力，一些研究已提出組合多個深度學習（DL）模型來提高檢測性能。Mirsky等人 [104] 提出了基於自組合（AE）的集成網絡入侵檢測系統（NIDS）Kitsune。Kitsune是一種完全無監督的方法，利用阻尼增量統計和自組合重建錯誤率來檢測潛在的異常流量。該方法計算網絡中每個通信通道的時間統計，然後將其映射並輸入到一組自組合中。每個自組合的重建錯誤由另一個全連接自組合進一步處理，以產生流量的最終異常分數。儘管Kitsune的完全無監督性使其更易於訓練和部署，但對時間特征的依賴使其易受規避攻擊。此外，該方法的完全無監督性進一步加劇了這種脆弱性。Nandanwar等人. [107] 最近提出了AttackNet，一種基於卷積神經網絡（CNN）和閘門迴歸單元（GRU）組合的模型，用於對網絡時間序列數據進行分類。所提出的架構利用有限數量的閘門來緩解過擬合。此外，GRU和CNN的組合使得能夠分析數據的時間和空間特徵，提高了集成模型的檢測能力。Hasan et Securing industrial internet of things against botnet attacks using hybrid deep learning approachal. [61] 通過結合長短期記憶網絡（LSTM）和DNN的混合深度學習策略來處理工業互聯網（IIoT）環境中的入侵檢測，

LSTM and DNN. An LSTM netowkr processes the time series sensor data, while a DNN separately processes non-time series traffic features. A single dense layer is used to combine the outputs of the LSTM and DNN and perform classification. Similarly, a recent study by Nazir et al. [108] explores the application of LSTM and CNN to anomaly detection in IoT networks. Initially, a CNN is used to learn and extract structural features of the data, followed by a pooling layer to reduce the dimensionality of these features. In parallel, the input data is fed to an LSTM network to identify temporal correlations. The outputs of the LSTM and CNN are combined through a dense layer to perform classification. All these approaches utilize a similar detection architecture, heavily relying on temporal correlations in the network traffic to identify malicious activity. While temporal relations are powerful features for classifying in-distribution data, their unconstrained nature makes them vulnerable to manipulation by attackers, allowing for easier evasion of detection. Yaser et al. [151] introduce a two-step architecture to detect DDoS attacks based on AE, CNN, and LSTM networks in SDN environments. A fully connected AE is used to learn descriptive features from the flow table entries and associated statistics of SDN switches. A CNN and an LSTM network are then trained on the labeled features extracted by the autoencoder to identify DDoS traffic. While unsupervised feature learning enables the extraction of meaningful features from the flow table entries, the use of unconstrained statistics makes the system more susceptible to evasion attacks. Furthermore, the reliance on easily manipulated features further exacerbates this vulnerability.

TAKAWAY. Ensembles of DL models exhibit excellent classification performance across a wide range of settings. The ability of DNNs to learn powerful, descriptive features — even from unlabeled data — and their capacity to generalize to unseen data make DL-NIDS highly versatile. Furthermore, utilizing ensembles of heterogeneous DL architectures allows for the effective exploitation of both spatial and temporal features, leveraging the specialized strengths of each architecture. However, the opaque nature of the feature learning and decision-making processes of DNNs reduces explainability, making it difficult to interpret how classification decisions are made. Moreover, directly applying DNNs to network-level features without incorporating architectural or data-level safeguards leaves the models highly vulnerable to adversarial attacks, as malicious inputs can be generated through a variety of means to exploit DNN's weaknesses.

4.3 Discussion

Extensive research has demonstrated the potential of both traditional ML and DL approaches in NIDS. Traditional ML approaches typically rely on expert knowledge for feature selection and engineering, which simplifies the learning tasks and can improve performance in specific settings. Moreover, their simpler nature allows for better explainability and improved confidence in classification results. On the other hand, DL-NIDS can autonomously learn complex, non-linear relations in the input data and typically generalize better to unseen queries compared to ML-NIDS. However, regardless of the specific learning algorithm used, robustness and resilience to adversarial attacks remain insufficiently addressed concerns. DNNs are heavily susceptible to gradient-based attacks such as adversarial examples (see Section 3.4). ML-NIDS are typically non-differentiable and, therefore, may appear robust against evasion attacks. However, adversaries can leverage differentiable approximations of the models to craft adversarial examples, and several specific adversarial attacks have been developed specifically for traditional ML techniques [23, 41, 154]. Although relying on ensembles generally enhances robustness against adversarial attacks, the use of multiple models significantly increases computational costs and classification time, limiting the practicality of these approaches in certain scenarios.

Ultimately, to develop NIDS that are resilient to adversarial attacks, it is crucial to thoroughly explore the trade-offs between explainability and modeling capability, as well as between robustness and computational complexity. Moreover, it is essential to understand how the structured nature of network traffic and interdependencies between features affect adversarial attacks. While combinations of ML and DL techniques combine the modeling capabilities of DL with the expert knowledge and explainability of traditional ML, significant gaps remain in

LSTM和DNN。一個LSTM網絡處理時間序列感應數據，而DNN則分別處理非時間序列流量特徵。使用單個密集層來組合LSTM和DNN的輸出並進行分類。類似地，Nazir等人. [108] 最近研究了LSTM和CNN在物聯網（IoT）網絡異常檢測中的應用。最初，使用CNN來學習和提取數據的結構特徵，然後通過池化層降低這些特徵的維度。同時，輸入數據被輸入到LSTM網絡中以識別時間相關性。LSTM和CNN的輸出通過密集層組合以進行分類。所有這些方法都利用了類似的檢測架構，嚴重依賴網絡流量中的時間相關性來識別惡意活動。雖然時間關係是分類分佈內數據的強大特徵，但其無約束性使得它們容易受到攻擊者的操縱，允許更容易規避檢測。Yaser等人. [151]介紹了一種基於自組合（AE）、CNN和LSTM網絡的兩步驟架構，用於在软件定義網絡（SDN）環境中檢測DDoS攻擊。一個全連接自組合用於從SDN交換機的流表條目和相關統計學中學習描述性特徵。然後，CNN和LSTM網絡在自組合提取的標記特徵上進行訓練以識別DDoS流量。雖然無監督特徵學習能夠從流表條目中提取有意義的特徵，但使用無約束統計學使得系統更容易受到規避攻擊。此外，對於容易操縱的特徵的依賴進一步加劇了這種脆弱性。

TAKAWAY。深度學習模型集合在廣泛的設定中展現出卓越的分類性能。深度神經網路學習強大、描述性特徵的能力——即使從未標記數據中——以及其泛化到未見數據的能力，使深度學習網路入侵偵測系統（DL-NIDS）高度靈活。此外，利用異構深度學習架構的集合有助於有效地利用空間和時間特徵，發揮每個架構的專門優勢。然而，深度神經網路特徵學習和決策過程的不透明性降低了可解釋性，使得難以解釋分類決策是如何做出的。此外，直接將深度神經網路應用於網路層特徵而不加入架構或數據層的保護措施，會使模型高度易受對抗攻擊，因為惡意輸入可以透過各種方式生成以利用深度神經網路的弱點。

4.3 討論

廣泛的研究已證明了傳統機器學習和深度學習方法在網路入侵偵測系統（NIDS）中的潛力。傳統機器學習方法通常依賴專家知識進行特徵選擇和工程，這簡化了學習任務並可以在特定設定中提高性能。此外，它們更簡單的性質有助於提高可解釋性並增強對分類結果的信心。另一方面，深度學習網路入侵偵測系統可以自主學習輸入數據中的複雜、非線性關係，並且通常比傳統機器學習網路入侵偵測系統更好地泛化到未見的查詢。然而，無論使用特定的學習算法如何，對抗攻擊的穩健性和韌性始終是未充分關注的問題。深度神經網路對基於梯度的攻擊（如對抗樣本，見第3.4節）非常易受攻擊。傳統機器學習網路入侵偵測系統通常是不可微分的，因此可能看起來對規避攻擊很穩固。然而，攻擊者可以利用模型的可微分近似來製造對抗樣本，並已開發出一些針對傳統機器學習技術的特定對抗攻擊 [23, 41, 154]。雖然依賴集合通常能增強對抗對抗攻擊的穩健性，但使用多個模型顯著增加了計算成本和分類時間，限制了這些方法在某些場景中的實用性。

最終來說，要開發能抵禦對抗性攻擊的NIDS，關鍵在於徹底探討解釋性與建模能力之間、以及穩健性與計算複雜性之間的權衡。此外，理解網路流量結構性質以及特徵間相互依賴如何影響對抗性攻擊也至關重要。雖然ML和DL技術的組合結合了DL的建模能力與傳統ML的專家知識和解釋性，但在文獻中仍存在顯著的差距，主要原因是對解釋性分析、實際應用場景和對抗性穩健性考慮不足。

the literature, mainly due to insufficient consideration for explainability analysis, real-world scenarios, and adversarial robustness. This raises open questions that remain unexplored:

Question 1. How effective are adversarial attacks against structured data such as network traffic? Can adversarial attack algorithms effectively account for interdependencies and constraints among features to generate realistic manipulations applicable in real-world scenarios?

Question 2. How well do ensemble methods resist adversarial attacks in comparison to single-learning approaches and adversarial training methods? Is the increased computational cost justified by the enhanced robustness provided?

The following sections critically review the literature on the vulnerabilities of intelligent NIDS to adversarial threats. By analyzing how current research replies to these questions, we aim to identify advancements and gaps for developing robust IDS able to counteract evolving adversarial strategies.

5 ANALYSIS OF ADVERSARIAL THREAT TO NETWORK INTRUSION DETECTION SYSTEMS

Evasion techniques, such as adversarial examples, present a fundamental threat to intelligent NIDS. However, unlike domains using unstructured data (e.g., image/audio classification, NLP), generating adversarial examples against NIDS is more challenging due to the interdependency and constraints between features in structured data. Network traffic follows specific protocols with predefined features, data types, and relations between different packet fields, making it difficult to manipulate the traffic coherently while also generating effective attacks. Adversarial manipulations must maintain the semantic validity of the data (e.g., changing the protocol field to a non-existent value) and its underlying structure (e.g., changing a packet header format). This section presents an analytical study of existing research on methods for generating adversarial examples that evade ML-NIDS, categorized into white-box, black-box, and gray-box attacks (summarized in Table 3). By examining the strengths and weaknesses of different approaches, we highlight the threat of adversarial attacks against NIDS and identify significant gaps in the current literature.

5.1 Generation of White-Box Attacks

In white-box attack settings, adversaries have full knowledge of the target NIDS, including architecture and model parameters. While such a level of access might be unlikely in real-world scenarios [13], white-box attacks provide valuable insights into potential attack vectors and vulnerabilities.

5.1.1 Vulnerability of DL-NIDS:. The application of traditional, gradient-based attacks against DL-NIDS has been analyzed by several studies in the literature. In [105], the authors analyze the vulnerability of a DNN-based NIDS against JSMA [115]. The authors explore all possible combinations of network traffic features and employ a modified saliency map to prioritize the features that minimize the true class probability. Effectively, through the modified saliency map, the attack determines the most sensitive feature combination to craft adversarial perturbations. Similarly, in [17] Ayub et al. study the vulnerability of MLP NIDS to JSMA under different training dataset. While they highlight the effectiveness of the attack, these studies do not consider the interdependencies and constraints between traffic features. This limits the applicability of the findings, as the authors do not verify whether the adversarial perturbations respect the semantic validity of the data, or whether they affect the expected behavior of the network traffic in a realistic scenario.

Pawlicki et al. [118] explored several adversarial attacks such as CW [28], FGSM [55], BIM [82], and PGD [91] against DNN-NIDS. All the adversarial attacks were grouped into a single attack class for the IDS training and testing. The results of the study highlight the effectiveness of different gradient-based adversarial attacks in causing misclassification. However, since the adversarial attacks were applied without modification, the generated adversarial perturbations did not respect the semantic validity of the network traffic (e.g., negative packet size). The authors work around this issue by zeroing out invalid fields. However, such an approach still does not

這引發了仍待探索的開放性問題：

問題 1。針對結構化數據（如網路流量）的對抗性攻擊有多有效？對抗性攻擊演算法能否有效考慮特徵之間的相互依存性和約束，以生成適用於實際場景的現實性操縱？

問題 2。與單一學習方法和對抗性訓練方法相比，集成方法抵抗對抗性攻擊的效果如何？增加的計算成本是否被提供的增強韌性所證明？

以下幾節批判性地回顧了有關智能 NIDS 對對抗性威脅的脆弱性的文獻。通過分析當前研究如何回應這些問題，我們旨在識別開發能夠抵禦演變的對抗性策略的韌性 IDS 的進步和差距。

5 對網路入侵檢測系統的對抗性威脅分析

迴避技術，如對抗性範例，對智能 NIDS 構成基本威脅。然而，與使用非結構化數據的領域（例如，圖像/音頻分類、NLP）不同，由於結構化數據中特徵之間的相互依存性和約束，對 NIDS 生成對抗性範例更具挑戰性。網路流量遵循特定協議，具有預設的特徵、數據類型和不同包字段之間的關係，這使得在生成有效攻擊的同時，難以協調地操縱流量。對抗性操縱必須保持數據的語義有效性（例如，將協議字段更改為不存在的值）及其底層結構（例如，更改包頭格式）。本節提出了一項分析研究，有關現有研究中的方法，以生成迴避 ML-NIDS 的對抗性範例，分類為白盒、黑盒和灰盒攻擊（總結於表 3）。通過檢查不同方法的優缺點，我們突顯了對 NIDS 的對抗性攻擊的威脅，並識別了當前文獻中的顯著差距。

5.1 白盒攻擊的產生

在白盒攻擊設定中，攻擊者對目標NIDS有完全的了解，包括其架構和模型參數。雖然這種程度的訪問在真實世界情況中可能不太可能， [13], 白盒攻擊能提供有價值的洞見，了解潛在的攻擊向量與漏洞。

5.1.1 DL-NIDS的漏洞：. 文獻中已有數項研究分析過傳統、基於梯度的攻擊對DL-NIDS的應用。在 [105], 該作者分析基於DNN的NIDS對JSMA的漏洞。作者探索了所有可能的網路流量特徵組合，並使用修改過的顯著性地圖來優先處理能最小化真實類別機率的特徵。實際上，透過修改過的顯著性地圖，攻擊能判斷最敏感的特徵組合來製造對抗性干擾。類似地，在 [17] Ayub等人研究不同訓練數據集下MLP NIDS對JSMA的漏洞。雖然他們強調了攻擊的有效性，這些研究並未考慮流量特徵之間的相互依存關係與限制。這限制了研究結果的應用性，因為作者並未驗證對抗性干擾是否尊重數據的語義有效性，或是否影響網路流量在真實情況下的預期行為。

Pawlicki 等人 [118] 探索了幾種對 DNN-NIDS 的對抗攻擊，例如 CW [28],FGSM [55],BIM [82], 和 PGD [91]。所有對抗攻擊都被分組歸入單一攻擊類別，用於 IDS 的訓練和測試。研究結果突顯了不同基於梯度的對抗攻擊在導致錯誤分類方面的有效性。然而，由於對抗攻擊是未經修改地應用的，因此生成的對抗擾動並未遵守網絡流量（例如，負包大小）的語義有效性。作者通過將無效字段置零來解決這個問題。然而，這種方法仍然無法

Table 2. Comprehensive summary of recent papers for the enhancement of machine learning and deep learning based NIDS

Ref.	Year	Feature Selection	Learning Method	Algorithms	Dataset	Pros	Cons
[34]	2020	Model-based importance score	Single learning	ML (DT & RF)	NSL-KDD & UNSW	Low false positives	Feature selection bias, lack of robust features
[4]	2020	Markov Chain	Single learning	ML (RF)	ISCX, Stratosphere IPS, CTU-13	Privacy-preserving, unseen detection	Botnet-only, weak to obfuscation/reordering evasion
[99]	2019	Manual (entropy)	Single learning	ML (SVM)	Mininet simulation	Transparent, integration with SDN controller	DoS-only, easily manipulated features
[160]	2018	DFEL	Single learning	DL (DFNN)	NSL-KDD & UNSW-NB15	Time-saving applications	Inadequate capacity to reduce data dimensionality
[92]	2020	DBN	Single learning	DL (DBN)	CICIDS 2017	Effective detection rates, potential for scalability	Evasion through feature deletion
[9]	2017	AE	Single learning	DL (DNN)	NSL-KDD, IBM NSCS & IBM TrueNorth	Potential for scalability	Overfitting
[83]	2021	PCA+Mutual Information (MI)	Single learning	DL (LSTM)	KDD'99	Great capability of learning	Overfitting, easily manipulated features
[121]	2021	AE	Single learning	DL (MLP)	ClamAV, Ember, GT Malware, MalRec, MalShare, VirusShare, VX Underground & Miscellaneous	High detection rate, robustness to noise	Sensitivity to advanced adversarial attacks
[79]	2021	Manual	Single learning	DL (CNN)	Public IoT data set created by [100]	Multi-type attacks detection	Biased feature selection
[138]	2023	Sequential Forward Selection (SFS) with statistical ranking	Single learning	DL (DNN)	NSL-KDD, UNSW-NB-15, & CIC-IDS-2017	Informative feature selection	Sensitivity to outliers
[94]	2018	DBN	Multimodel learning	DL & ML	KDD'99 CUP, NSL-KDD, UNSW-NB15, & CICIDS2017	Robust to imbalanced data	Sensitivity to outliers
[74]	2022	Non-symmetric deep AE	Multimodel learning	DL & ML (stacked (NDAE)& SVM)	KDD CUP'99	Less susceptible to overfitting	Missing subtle features
[122]	2022	Implicit feature learning	Multimodel learning	DL (RNN, LSTM, & GRU) & ML (RF, SVM, & LR)	KDD-Cup'99, SDN-IoT, UNSW-NB15, WSN-DS, & CICIDS-2017	Automated feature learning, capturing temporal patterns	Overfitting, difficult interpretation
[135]	2019	Hybrid (Evolutionary algorithms+REP Tree)	Multimodel learning	ML (Rotation Forest & bagging)	NSL-KDD & UNSW-NB15	Robust feature selection	Sensitivity to outliers
[159]	2020	Hybrid (CFS+Evolutionary approach)	Multimodel learning	ML (C4.5, RF, & Forest by PA)	NSL-KDD, AWID, & CIC-IDS2017	Robust feature selection	Sensitivity to feature correlations
[58]	2019	Feature augmentation	Multimodel learning	ML (DT- EnSMV)	NSL-KDD	Handle the complex feature interdependencies	Limited feature control
[14]	2024	XAI & Information gain	Multimodel learning	DL & ML (DNN, DT, KNN, MLP, AdaBoost, Logistic Regression, SVM, & Extreme Gradient Boosting)	RoEduNet-SIMARGL2021, NSL-KDD, & CICIDS-2017	Improved individual and ensemble performance	Limited outlier detection
[107]	2024	Implicit feature learning	Multimodel learning	DL (CNN & GRU)	N_BaIoT	Less susceptible to overfitting, simple architecture	Sensitivity to outliers, missing important features
[61]	2022	Implicit feature learning	Multimodel learning	DL (LSTM & DNN)	N_BaIoT	Scalability and generalizability for handling large datsets	Limited feature control, sensitivity to noisy data
[108]	2024	Implicit feature learning	Multimodel learning	DL (CNN & LSTM)	N_BaIoT & CICIDS2017	Scalability and generalizability	Overfitting, sensitivity to outliers
[151]	2022	AE	Multimodel learning	DL (CNN & Bidirectional Long Short-Term Memory)	ISCX-IDS-2012 & UNSW2018	Improved DDoS attack detection	Overfitting, misinterpretation by CNNs and BDLSTMs, sensitivity to outliers

guarantee the semantic validity of network packets, potentially resulting in perturbed features that cannot be achieved in a real network setting. Furthermore, the authors do not discuss feature interdependencies and do not consider feature constraints in their attacks. In [62], Hashemi et al. study how correlations between network

表2. 為提升機器學習與深度學習基礎的NIDS所做近期論文的綜合總結

Ref.	Year	特徵選擇	學習方法	演算法	資料集	Pros	Cons
[34]	2020	基於模型的單一實例學習	分數學	機器學習 (決策樹 & 隨機森林)	NSL-KDD & UNSW	低偽陽性率	特徵選擇偏差，缺乏穩健性特色
[4]	2020	馬可夫鏈	單一學習	ML (RF)	ISCX、Stratosphere IPS、CTU-13	隱私保護，未見檢測tion	僅Botnet， weak to 混淆遮蔽/重排迴避
[99]	2019	手動 (熵)	單一學習	ML (SVM)	Mininet 模擬	透明，與 SDN 整合控制器	DoS 僅，易受操縱的特徵特徵
[160]	2018	DFEL	單一學習	DL (DFNN)	NSL-KDD & UNSW-NB15	節省時間的應用	減少數據的容量不足維度
[92]	2020	DBN	單一學習	DL (DBN)	CICIDS 2017	有效偵測率、潛在透過特徵刪除實現可擴展性	
[9]	2017	AE	單一學習	DL (DNN)	NSL-KDD、IBM NSCS & IBM TrueNorth	可擴展性潛力	過度擬合
[83]	2021	PCA+互信息(MI)學習		DL (LSTM)	KDD'99	學習能力強	過擬合，輕易被操縱的特徵
[121]	2021	AE	單一學習	DL (MLP)	ClamAV、Ember、GT Malware、MalRec、MalShare、VirusShare、VX 地下&無類	高偵測率，對噪音	對先進的對抗性攻擊的敏感度攻擊
[79]	2021	手動	單一學習	DL (CNN)	公開 IoT 資料集 created by [100]	多類型攻擊偵測	偏見特徵選擇
[138]	2023	序列前向單選擇 (SFS) 與學習統計排名		DL (DNN)	NSL-KDD、UNSW-NB-15、& CIC-IDS-2017	資訊量化的特徵選擇	對離群值的敏感性
[94]	2018	DBN	多模型學習	DL & ML	KDD'99 CUP、NSL-KDD、UNSW-NB15、& CICIDS2017	對不平衡數據的穩定性	對離群值的敏感度
[74]	2022	非對稱深度AE	多模型學習	DL學習 & ML (堆疊(NDAE)& SVM)	KDD CUP'99	較不容易過度擬合	錯失微妙的特徵
[122]	2022	隱含的特性學習	多模型學習	深度學習 (RNN、LSTM、& GRU) & 機器學習 (RF、SVM、& LR)	KDD-Cup'99， SDN-IoT，UNSW-NB15， WSN-DS，& CICIDS-2017	自動特徵學習，捕捉時間模式ing 時間模式	過度擬合，難以解釋
[135]	2019	混合 (演進-式演算法+REP樹)	多模型學習	機器學習 (旋轉森林 & 袋狀法)	NSL-KDD & UNSW-NB15	強健特徵選擇	對離群值的敏感性
[159]	2020	混合 (CFS+演進方法)	多模型學習	機器學習 (C4.5, 隨機森林，& 森林由PA)	NSL-KDD，AWID，& CIC-IDS2017	強健特徵選擇	對特徵相關性的敏感性
[58]	2019	功能增強tion	多模型學習	ML (DT- EnSMV)	NSL-KDD	處理複雜的特徵互動依賴性	有限功能控制
[14]	2024	XAI & 資訊gain	多模型深度學習 & 机器学习 (DNN, DT,KNN, 学习	MLP, AdaBoost, 逻辑回归回归, SVM, & 极端梯度提升)	RoEduNet-SIMARGL2021, NSL-KDD、& CICIDS-2017	改進個別與合奏的限外檢測性能	
[107]	2024	隱含學習 特性	多模型學習	DL (CNN & GRU)	N_BaIoT	較不易過擬合，簡單- 對異常值、缺失重要特徵敏感的架構	
[61]	2022	隱含學習 特性	多模型學習	深度學習 (LSTM & DNN)	N_BaIoT	處理大型數據集的可擴展性和泛化能力	有限的功能控制，對雜訊數據敏感
[108]	2024	隱式學習 功能	多模型學習	DL (CNN & LSTM)	N_BaIoT & CICIDS2017	可擴展性與泛化能力	過擬合、對離群值的敏感性
[151]	2022	AE	多模型學習	DL(CNN &雙向長短期記憶 ISCX-IDS-2012)	& UNSW2018	改善DDoS攻擊偵測	過度擬合、CNNs和BDLSTMs的誤解讀、對離群值的敏感性

保證網路封包的語義有效性，可能導致在真實網路環境下無法達成的干擾特徵。此外，作者們沒有討論特徵之間的相互依存關係，也沒有在他们的攻擊中考慮特徵約束。在 [62], Hashemi等人研究中，他們研究網路之間的相關性

traffic features impact the effectiveness of adversarial attacks. The authors evaluate adversarial attacks against two types of DL-NIDS:

- *Packet-based NIDS:* The proposed attack relies on a local copy of the target NIDS to identify packet-level perturbations that lead to misclassification. The authors define a set of legitimate packet transformations that maintain the semantic validity of the data. Differently from previous works, the authors do not utilize gradient-based techniques but rely on splitting, delaying, and injecting packets in the network.
- *Flow-based NIDS:* To account for feature interdependencies and constraints at the flow level, the authors categorize the features into four groups based on how freely each feature can be perturbed while preserving consistency. During the attack, three distinct masks are employed to determine which group of features to alter. The attack proceeds through multiple iterations of perturbations, progressively expanding the permissible range of alterations to generate the adversarial examples.

Searching for an adversarial perturbation with a lower score can create a buffer zone between perturbations and real-world traffic, especially for dependent features outside the attacker's control. However, variations in real-world attack traffic might cause the NIDS score to fall outside this buffer zone, leading to detection. This highlights the challenge of accurately simulating the dynamic properties of real-world network traffic.

5.1.2 Vulnerability of Traditional ML-NIDS:. Several attacks have been proposed against Traditional ML-NIDS. Anthi et al. [11] studied the effectiveness of JSMA against ML-NIDS based on RF and J48 classifiers in ICS environments. Since RF and J48 are non-differentiable algorithms and JSMA requires gradient computation, the authors rely on a pre-trained DNN that is used as a proxy to generate the adversarial perturbations instead of the target the ML-NIDS. While utilizing a differentiable proxy model to generate the adversarial attacks shows promising results, the authors do not address concerns related to the transferability of the attacks to the original models, nor address the problem of feature consistency and semantic validity of the perturbed samples.

In another study, Alhajjar et al. [7] investigated the sensitivity of different ML-NIDS (e.g., bagging, gradient boosting, linear discriminant analysis, and quadratic discriminant analysis) to adversarial manipulations. The authors used Genetic Algorithm (GA) and Particle Swarm Optimization (PSO) to iteratively modify malicious traffic data and employed a Generative Adversarial Network (GAN) to generate adversarial samples. The GAN's generator generates adversarial samples, while the discriminator tries to distinguish them from real traffic. To respect feature constraints, the authors relied on expert knowledge to identify features that may be altered without disrupting network functionality. While the proposed approach shows promising results and respects feature constraints, feature interdependencies and semantic validity are not taken into consideration. Moreover, the proposed approach is computationally expensive, limiting its applicability to real-time attacks.

TAKEAWAY. While several studies confirm the vulnerability of intelligent NIDS to white-box adversarial attacks, these works frequently ignore crucial real-world complexities such as feature interdependency, feature constraints, and semantic validity of the perturbed samples, highlighting inconsistencies with real-world threat model settings. Moreover, in the case of ML-NIDS, the transferability of adversarial attacks crafted against proxy models to the original ML-NIDS remains uncertain. Further research is required to fully assess the extent of this threat.

5.2 Generation of Black-Box Attacks

Black-box attack settings assume that the attacker has no knowledge of the details of the target NIDS, such as architecture or parameters. Compared to white-box attacks, black-box settings provide a more realistic threat model and are more representative of real-world scenarios. Typically, black-box attacks rely on an "oracle" model that adversaries can query to validate whether the NIDS detects the perturbed traffic. This feedback from the oracle model is utilized to further refine the perturbation until it achieves evasion. Figure 4 illustrates a general framework for these attacks.

交通特徵影響了對抗性攻擊的有效性。作者們評估了針對兩種類型的 DL-NIDS 的對抗性攻擊：

- 基於封包的 NIDS：所提出的攻擊依賴於目標 NIDS 的本地副本，以識別導致錯誤分類的封包層級干擾。作者們定義了一組保持數據語義有效性的合法封包變換。與先前的工作不同，作者們不利用基於梯度的技術，而是依賴於在網路中分割、延遲和注入封包。
- 基於流量的 NIDS：為了考慮特徵之間的相互依存性和流量層級的約束，作者們根據每個特徵在保持一致性時可以自由變動的程度將其分為四組。在攻擊過程中，採用三個不同的掩碼來確定要改變哪一組特徵。攻擊通過多次變動迭代進行，逐步擴大允許的變動範圍，以生成對抗性範例。

尋找一個分數較低的對抗性變動可以創造一個變動和真實世界交通之間的緩衝區，特別是對於攻擊者無法控制的相依特徵。然而，真實世界攻擊交通的變化可能導致 NIDS 分數超出這個緩衝區，導致被偵測。這突顯了準確模擬真實世界網路交通動態特性的挑戰。

5.1.2 傳統機器學習網路入侵偵測系統的漏洞：已有數種攻擊方法針對傳統機器學習網路入侵偵測系統提出。Anthi 等人 [11] 研究了 JSMA 對於基於 RF 和 J48 分類器的工業控制系統環境中的 ML-NIDS 的有效性。由於 RF 和 J48 為非可微算法，且 JSMA 需要梯度計算，作者們依賴一個預訓練的 DNN 作為代理來生成對抗性干擾，而不是目標 ML-NIDS。雖然利用可微代理模型生成對抗性攻擊顯示出有希望的結果，但作者們未解決攻擊對原始模型的轉移性相關問題，也未解決干擾樣本的特徵一致性與語義有效性問題。

在另一項研究中，Alhajjar 等人 [7] 調查了不同 ML-NIDS（例如，bagging、梯度提升、線性判別分析以及二次判別分析）對於對抗性操作的敏感性。作者們使用遺傳算法 (GA) 和粒子群優化 (PSO) 來迭代修改惡意流量數據，並採用生成對抗性網絡 (GAN) 來生成對抗性樣本。GAN 的生成器生成對抗性樣本，而判別器嘗試將其與實際流量區分開來。為了尊重特徵約束，作者們依賴專家知識來識別可能被修改而不會中斷網絡功能的特徵。雖然所提出的方法顯示出有希望的結果並尊重特徵約束，但特徵間的相互依賴性和語義有效性並未被考慮。此外，所提出的方法計算成本高昂，限制了其在實時攻擊中的應用性。

TAKEAWAY。雖然有多項研究確認智慧型NIDS易受白盒對抗攻擊，但這些研究經常忽略真實世界中的關鍵複雜性，例如特徵間相依性、特徵限制，以及干擾樣本的語義有效性，這凸顯了其與真實世界威脅模型設定之間的不一致性。此外，在ML-NIDS的情況下，針對代理模型製造的對抗攻擊對於原始ML-NIDS的轉移性仍然不明確。需要進一步研究才能全面評估這項威脅的程度。

5.2 黑盒攻擊的生成

黑色盒攻擊設定假設攻擊者對目標NIDS的細節無知，例如架構或參數。與白盒攻擊相比，黑色盒設定提供更現實的威脅模型，更能代表真實世界的情況。通常，黑色盒攻擊依賴於一個「神祕計算」模型，敵人可以向該模型查詢以驗證NIDS是否偵測到干擾的流量。這個神祕計算模型的回饋被用來進一步精煉干擾，直到達成迴避。圖4說明了這些攻擊的通用框架。

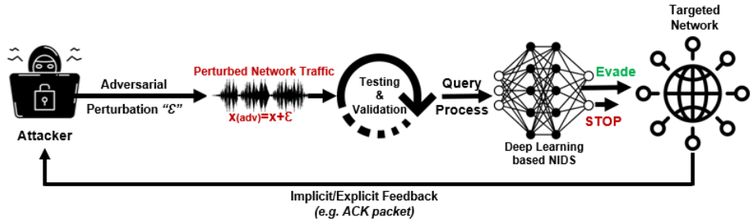


Fig. 4. Adversarial perturbation against Neural Network Based-NIDS *(Black-Box)*

5.2.1 Vulnerability of DL-NIDS:. In [153], the authors evaluate C-LSTM, an ensemble model combining CNN and LSTM, and individual DNN and CNN models against a variety of black-box attacks: Natural Evolution Strategies (NES) [73], Boundary Attack [20], HopSkipJumpAttack [30], Pointwise Attack [128], and OPT-Attack [88]. The authors assessed various aspects of adversarial sample generation, including attack success rate, query efficiency, and transferability. The results of the study highlight the effectiveness of these attacks against both single-model and ensemble NIDS. However, the study does not properly address concerns related to feature interdependencies and constraints. Moreover, crucial details of the attacks related to feature selection and the query process were not covered, making it difficult to evaluate the real-world applicability of the findings. In [119], the authors explore the creation of adversarial DoS attacks aimed at evading DNN-NIDS. Their analysis reveals that traditional constraints for generating adversarial examples, which rely on distance metrics such as Lp norms, are ineffective when used for DoS traffic. These metrics treat all features equally, despite DoS traffic containing interconnected features with highly variable value ranges. To address this limitation, the authors utilize Mahalanobis distance, which accounts for feature scale differences, enabling more accurate similarity measures between normal and adversarial DoS traffic. While employing different constraints on perturbations helps to better capture individual feature constraints, the approach still overlooks feature interdependencies and semantic validity of the perturbed traffic. Wu et al. [146] propose a deep reinforcement learning approach consisting of a black-box botnet NIDS and a flow manipulation module interacting with a Deep Q-Learning agent. For each network flow, the agent selects a modification action from a predetermined set that changes communication patterns without affecting functionality. The modified flow is evaluated by the detection model, and the agent is rewarded based on whether or not it is detected as malicious. This feedback loop enables the agent to iteratively refine the learned perturbation strategy and maximize evasion. While relying on a predefined set of modifications ensures that feature constraints and interdependencies are satisfied, it significantly restricts the perturbation space that the agent can explore, thereby limiting the overall effectiveness of the attack.

In [129], the authors propose TANTRA, an evasion attack aimed at models relying on temporal correlations between packets. An LSTM model is used to learn inter-packet arrival times for benign traffic. Then, malicious traffic is generated following the learned pattern, simulating the temporal characteristics of benign traffic without altering the packets' content. While TANTRA can effectively manipulate temporal features to achieve evasion, most NIDS combine timing statistics with other packet or flow information to detect malicious traffic. Consequently, the applicability of this evasion attack is generally limited, as it only targets a subset of the features that are typically used for network intrusion detection. Debicha et al. [44] focus on evading flow-based NIDS targeted at botnets. Unlike previous works, the authors train a surrogate model on benign traffic as a proxy for the real NIDS and generate adversarial perturbations relying on the output of the surrogate model. The authors ensure that feature constraints and semantic validity are satisfied through feature masking and projection

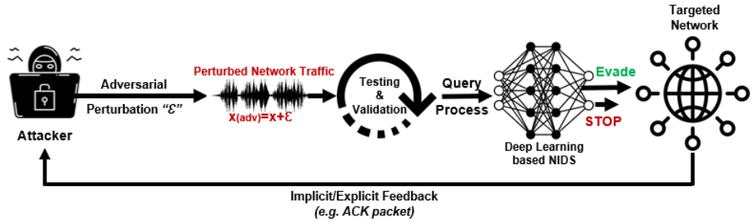


圖 4. 對基於神經網絡的網路入侵偵測系統（黑盒）的對抗性干擾

5.2.1 DL-NIDS的漏洞：在 [153], 中，作者評估了結合CNN和LSTM的集成模型C-LSTM，以及個別的DNN和CNN模型，對於各種黑盒攻擊：自然進化策略（NES） [73], 邊界攻擊 [20], HopSkipJumpAttack [30], 點攻擊 [128], 和OPT-Attack [88]。作者評估了對抗樣本生成的各個方面，包括攻擊成功率、查詢效率和可遷移性。研究結果突顯了這些攻擊對單一模型和集成NIDS的有效性。然而，研究沒有正確處理與特徵間相依性和約束相關的擔憂。此外，與特徵選擇和查詢過程相關的攻擊關鍵細節沒有涵蓋，使得難以評估研究結果的實際應用性。在 [119], 中，作者探討了旨在規避DNN-NIDS的對抗DoS攻擊的創建。他們的分析揭示，傳統的生成對抗樣本的約束，依賴於距離度量，如Lp範數，在用於DoS流量時是無效的。這些度量將所有特徵視為同等，儘管DoS流量包含具有高度變動值範圍的相互關聯特徵。為了解決這個限制，作者利用馬哈拉諾比斯距離，它考慮了特徵尺度差異，從而能夠在正常和對抗DoS流量之間進行更準確的相似性度量。雖然對擾動施加不同的約束有助於更好地捕捉個別特徵約束，但该方法仍然忽略了特徵間相依性和擾動流量的語義有效性。Wu等人 [146] 提出了一種深度強化學習方法，由一個黑盒僵尸網NIDS和一個與深度Q學習代理互動的流量操縱模塊組成。對於每個網絡流，代理從一個預定的集合中選擇一個修改動作，該動作改變通信模式而不影響功能。修改後的流被檢測模型評估，並且根據它是否被檢測為惡意來給予代理獎勵。這個反饋迴路使代理能夠迭代地改進學習的擾動策略並最大化規避。雖然依賴預定的一組修改確保了特徵約束和相依性得到滿足，但它顯著限制了代理可以探索的擾動空間，從而限制了攻擊的整體有效性。

在 [129], 中，作者提出 TANTRA，这是一种針對依賴封包間時間相關性的模型的迴避攻擊。使用 LSTM 模型來學習良性流量中封包的到達時間。接著，根據學習到的模式生成惡意流量，模擬良性流量的時間特性，而不改變封包內容。雖然 TANTRA 可以有效地操縱時間特徵以達成迴避，但大多數 NIDS 會結合時間統計資料與其他封包或流量資訊來偵測惡意流量。因此，這種迴避攻擊的應用性通常有限，因為它只針對網路入侵偵測通常使用的特徵子集。Debicha 等人. [44] 專注於迴避針對 botnet 的基於流量的 NIDS。與先前的研究不同，作者在良性流量上訓練一個代理模型作為真實 NIDS 的代理，並根據代理模型的輸出生成對抗性干擾。作者透過特徵遮罩和投影確保滿足特徵約束和語義有效性

operations. Feature masking allows perturbation only of features that can be freely modified (e.g., total number of packets), while the projection ensures that the domain of each feature is respected. The attack relies on the transferability of the adversarial attack between the surrogate model and the target NIDS. Similarly, Zolbayar et al. [161], developed NIDSGAN, a GAN-based attack that relies on a surrogate model to generate the adversarial perturbation. The GAN is trained using the surrogate model as a discriminator, and limitations on the samples generated by the GAN are imposed to maintain the semantic validity of the traffic. While using a surrogate model simplifies the generation of the adversarial perturbation, the effectiveness of the attack largely depends on how closely the decision boundary of the surrogate model resembles the original one. Moreover, constraining the features that can be modified reduces the overall perturbation space, potentially limiting the effectiveness of the attack by restricting its ability to find optimal adversarial examples.

5.2.2 Vulnerability of Traditional ML-NIDS:. Similar to [161], IDSGAN [86] utilizes a GAN architecture and a surrogate model to generate adversarial traffic. The surrogate model is employed as a distinguisher for the GAN training, providing a differentiable approximation of the target ML-NIDS. The proposed attack relies on a restricted modification mechanism to ensure that only non-functional features are altered for each type of attack, guaranteeing feature constraints and semantic validity. While the evaluation of the approach shows promising results, it is based on the simple and outdated NSL-KDD dataset, which does not accurately reflect modern real-world network traffic. Furthermore, limiting the allowed perturbations to non-functional features further restricts the attack's effectiveness.

TAKEAWAY. Existing studies suggest that Black-box attacks pose a significant threat to NIDS, regardless of their underlying architecture. However, the full extent of this threat remains uncertain, as many studies fail to account for real-world limitations in their threat models. Furthermore, most studies only partially address feature constraints and semantic validity of the perturbed traffic. These limitations underscore the need for further research to enhance NIDS defenses against evolving black-box attack strategies.

5.3 Generation of Gray-Box Attacks

In gray-box threat models, the attacker has partial knowledge of the NIDS features and architecture, allowing adversarial manipulations that are better targeted against the specific NIDS compared to the black-box setting.

Assuming the attacker knows the feature set used by the target NIDS, Vitorino et al. [144] propose the Adaptive Perturbation Pattern Method (A2PM). A2PM generates adversarial examples by analyzing features to identify valid ranges and correlations. It applies interval patterns to perturb numerical features within limits and combination patterns to preserve coherence in categorical features. These patterns are sequenced to introduce perturbations that comply with domain rules and class-specific requirements. Sheastsley et al. [130] propose two approaches to generate adversarial attacks that can adapt to specific network constraints:

- *Adaptive JSMA (ASJMA):* AJSMA incorporates network traffic constraints into JSMA. The attack relies on the concept of primary features, which are features that, when set, limit the permissible range of other features. When generating perturbations, AJSMA verifies whether a given feature is affected by another primary feature and limits the allowed perturbation according to their specific interdependencies.
- *Histogram Sketch Generation (HSG):* HSG learns meta-perturbations, called *sketches*, from a set of previously generated perturbations. HSG utilizes the results of previous attacks to detect frequently perturbed features and their modifications to learn a generalized attack pattern. The learned sketch can then be applied to new network data to evade detection.

While these techniques result in perturbations that satisfy feature constraints and interdependencies, they require extensive knowledge of features and target network traffic distribution, which may be limiting in real-life settings. In [78], the authors focus only on evaluating the robustness of ML-based ensembles against various well-known adversarial attacks. To enable the computation of gradient-based attacks on non-differentiable ML models, the

操作。特徵遮罩允許僅修改可以自由修改的特徵（例如，封包總數），而投影確保每個特徵的定義域得到尊重。這種攻擊依賴於對抗性攻擊在代理模型和目標 NIDS 之間的轉移性。類似地，Zolbayar 等人. [161], 開發了 NIDSGAN，這是一種基於 GAN 的攻擊，它依賴於代理模型來生成對抗性干擾。使用代理模型作為判別器來訓練 GAN，並對 GAN 生成的樣本施加限制，以維持流量的語義有效性。雖然使用代理模型簡化了對抗性干擾的生成，但攻擊的有效性很大程度上取決於代理模型的決策邊界與原始模型的相似程度。此外，限制可以修改的特徵減少了整體干擾空間，可能通過限制其尋找最佳對抗性範例的能力來限制攻擊的有效性。

5.2.2 傳統 ML-NIDS 的弱點：與 [161], IDSGAN [86] 一樣，採用 GAN 架構和替代模型來生成對抗流量。替代模型用於 GAN 訓練的區分器，為目標 ML-NIDS 提供可微分的近似。所提出的攻擊依賴於一種受限的修改機制，以確保每種攻擊類型僅修改非功能性特徵，保證特徵約束和語義有效性。雖然該方法的評估顯示出有希望的結果，但它基於簡單且過時的 NSL-KDD 資料集，這個資料集無法準確反映現代實際網絡流量。此外，將允許的干擾限制為非功能性特徵，進一步限制了攻擊的有效性。

TAKEAWAY。現有研究表明，黑盒攻擊對 NIDS 構成顯著威脅，無論其底層架構如何。然而，這種威脅的完整程度仍然不明確，因為許多研究未能在其威脅模型中考慮實際限制。此外，大多數研究僅部分處理了干擾流量的特徵約束和語義有效性。這些限制突顯了進一步研究以增強 NIDS 防禦對於演變的黑盒攻擊策略的必要性。

5.3 灰盒攻擊的生成

在灰盒威脅模型中，攻擊者對NIDS的特性與架構有部分知識，這使得針對特定NIDS的攻擊比黑盒設定下更具針對性。

假設攻擊者知道目標NIDS使用的特性集，Vitorino等人 [144] 提出了適應干擾模式方法（A2PM）。A2PM透過分析特性來識別有效範圍和相關性，生成對手樣本。它將區間模式應用於限制內的數值特性干擾，並將組合模式應用於保持類別特性的協調性。這些模式按序排列，以引入符合領域規則和類別特定需求的干擾。Sheastsley等人 [130]提出了兩種方法來生成能夠適應特定網路限制的對手攻擊：

- 適應性JSMA（ASJMA）：AJSMA將網路流量限制整合進JSMA。該攻擊依賴於主要特性的概念，這些特性在設置時會限制其他特性的可接受範圍。
  特性。在產生擾動時，AJSMA 會驗證給定特性是否受另一個主要特性影響，並根據其特定的相互依存關係限制允許的擾動。
- 直方圖草圖生成 (HSG)：HSG 從先前產生的擾動集合中學習元擾動，稱為草圖。HSG 利用先前攻擊的結果來檢測經常被擾動的特性
  及其修改，以學習一個泛化的攻擊模式。學習到的草圖可以應用於新的網絡數據以逃避免檢測。

雖然這些技術產生的擾動能滿足特徵約束和相互依存關係，但它們需要廣泛的特徵知識和目標網絡流量分佈，這在實際應用中可能存在限制。在 [78], 中，作者們僅關注評估基於機器學習的集成模型對各種著名對抗性攻擊的穩定性。為了實現對非可微分的機器學習模型的基於梯度的攻擊計算，

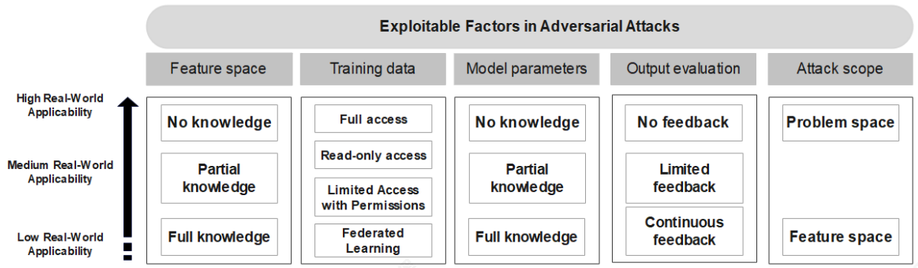


Fig. 5. Real-world applicability of adversarial factors when generating adversarial attacks.

authors employ a surrogate DNN model. The adversarial perturbation is computed using the surrogate model, with the success of the attack depending on the transferability of the generated perturbations to the target NIDS. While this paper approved the vulnerability of shallow models to adversarial samples, the original accuracy of the latter has not significantly dropped compared to other existing attacks (e.g., [7]). Additionally, further information about how the methodology considers the attacker's knowledge of the feature constraints or the data collected during the attack is not provided. Hence, this study does not reflect real-world conditions.

TAKEAWAY. Gray-box attacks strike a balance between the high complexity of black-box attacks and the limited real-world applicability of white-box attacks, posing a significant threat to intelligent NIDS. However, questions remain about their effectiveness in practical settings and the challenge of balancing attacker knowledge with realistic adversarial perturbations. Further research is required to accurately assess the risk posed by grey-box attacks and develop robust countermeasures for NIDS.

5.4 Discussion

This analytical review highlights significant gaps in the literature concerning the practical applicability of proposed adversarial approaches in network traffic. Several studies fail to account for feature constraints and interdependencies in structured data like network traffic. Furthermore, methodological details on how the attacker can gather relevant information for attacks, such as how to obtain a suitable surrogate model in black-box settings, are often lacking. Finally, several black-box proposals heavily rely on oracle access to the target NIDS, which is an unlikely proposition in real-world settings. These limitations underscore a significant gap between theoretical analysis and recommendations, and their practical feasibility.

The practical effectiveness of adversarial attacks depends on the attacker's access and knowledge of several key factors [13], as highlighted in Figure 5:

- **Feature space:** What level of knowledge does the attacker possess regarding the features utilized by the NIDS for classification?
- **Training data Access:** Is the training data for the ML-NIDS readable, modifiable, or completely inaccessible by the attacker?
- **Model parameters:** What level of knowledge does the attacker possess regarding the model architecture and parameters used by the NIDS?
- **Output evaluation:** Does the attacker have oracle access to the target NIDS, and what restrictions exist on the number of queries they can make?

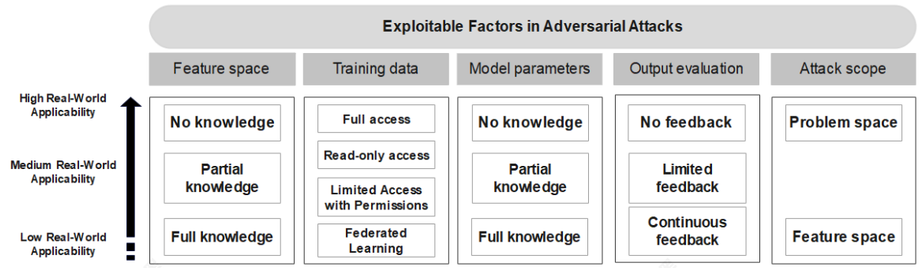


圖 5. 實際應用中的對抗因素，在產生對抗式攻擊時的應用性。

作者們採用了一個替代的深度神經網絡模型。對抗性擾動是使用替代模型計算的，攻擊的成功取決於生成的擾動對目標入侵檢測系統的轉移性。雖然這篇論文認可了淺層模型對對抗性樣本的易受性，但後者的原始準確率與其他現有攻擊（例如， [7]）相比並沒有顯著下降。此外，關於方法論如何考慮攻擊者對特徵約束的了解或攻擊過程中收集的數據的更多信息並未提供。因此，這項研究並不能反映真實世界的情況。

重點：灰盒攻擊在黑盒攻擊的高複雜性與白盒攻擊的實際應用性有限之間取得平衡，對智慧型 NIDS 構成重大威脅。然而，其實際環境中的有效性問題以及攻擊者知識與真實對抗干擾之間的平衡挑戰仍存疑。需要進一步研究以精確評估灰盒攻擊所造成的風險，並為 NIDS 開發堅固的對抗措施。

5.4 討論

這份分析性回顧突顯了文獻中關於所提出對抗方法在網路流量實際應用性方面的顯著差距。幾項研究未能考慮結構化數據（如網路流量）中的特徵限制與相互依存關係。此外，關於攻擊者如何收集攻擊相關資訊的方法論細節，例如如何在黑盒設定中獲得適當的替代模型，往往缺乏說明。最後，幾項黑盒建議高度依賴對目標 NIDS 的神祕訪問權限，而在實際環境中這並非可行。這些限制突顯了理論分析與建議之間的顯著差距及其實際可行性。

對抗性攻擊的實際效果取決於攻擊者對於幾個關鍵因素的接觸和了解 [13], ，如圖5所示：

- Featurespace：攻擊者對於NIDS（網路入侵偵測系統）用於分類的特徵，擁有何種程度的知識？
- 訓練資料存取：攻擊者能否閱讀、修改訓練ML-NIDS的資料，或完全無法存取？
- 模型參數：攻擊者對於NIDS使用的模型架構和參數，擁有何種程度的知識？
- 輸出評估：攻擊者是否有權限存取目標NIDS的「神諭」，以及對於他們可以發出的查詢數量有何限制？

Table 3. Summary of adversarial attacks on NIDS

References	Year	Type of Attack			Applicability		Limitation
		White-Box	Black-box	Gray-Box	Target Model	Dataset	
[105]	2023	✓	✗	✗	MLP	CIC-IDS2017, CIC-IDS2018, CIC-DDoS2019	No consideration of feature interdependency, lack of realism
[17]	2020	✓	✗	✗	MLP	CICIDS 2017, TRAbID 2017	No consideration of feature interdependency, lack of realism
[117]	2020	✓	✗	✗	DNN	CICIDS 2017	No consideration of feature interdependency Introduction of negative values into certain network traffic features
[62]	2019	✓	✗	✗	DNN	CICIDS 2017	Generalizability and lack of realism
[11]	2021	✓	✗	✗	RF, J48	Power System	Lack of details on handling feature correlations Limited generalizability due to pre-trained model Difficulty in balancing feature modification
[7]	2021	✓	✗	✗	SVM, RF, DT, NB, MLP, KNN, LR, GB, Bagging, LDA, QDA	NSL-KDD, UNSW-NB15	Impact on real-time attacks
[153]	2022	✗	✓	✗	C-LSTM,MLP, CNN	CSE-CICIDS 2018	No explanation on feature selection and correlations Limited information on the attacker's query process
[119]	2019	✗	✓	✗	DNN	KDDcup99, CICIDS2017	Limited effectiveness of traditional distance metrics Lack of details on the optimal settings for Mahalanobis distance parameters Lacks of details on how the attacker gathers information about the target model
[146]	2019	✗	✓	✗	CNN & DT	CTU13	Limited action space exploration Inability to directly exploit feature correlations
[44]	2023	✗	✓	✗	DNN	Ad-Hoc	Subtle, detectable perturbations Lack of explanation on how the attacker gathers information
[161]	2022	✓	✓	✗	DNN	NSL-KDD, CICIDS 2017	Using limited data for the surrogate model, reduce the attack success rates
[86]	2022	✗	✓	✗	SVM, MLP, KNN, RF, LR	NSL-KDD	Outdated data, lack of realism
[129]	2022	✗	✓	✗	LSTM	CIC-IDS2017, Kitsune	Complex real-time setup (less flexible)
[144]	2022	✗	✗	✓	MLP, RF	CIC-IDS2017, IoT-23	If the training data used by A2PM is inadequate, the generated attacks will be unrealistic and detected
[78]	2022	✓	✓	✓	AdaBoost, Bagging, Gradient Boosting, LR, DT, RF, SVC	NSL-KDD, UNSW-NB 15	No consideration of feature interdependency, lack of realism
[130]	2022	✓	✗	✓	ML *(Not Specified)*	NSL-KDD, UNSW-NB 15	Outdated data, lack of realism

- **Attack scope:** In what space are the adversarial perturbations generated? In the problem space (network traffic) or in the features space (learned model features)?

Further research in the area of robust NIDS should carefully account for the specific type of network being targeted, as well as the expected level of access and knowledge available to the adversary. The threat model considered should be designed to reflect these considerations, and the adversarial attacks used in the analysis must align with the level of access outlined in the threat model.

6 ANALYSIS OF PROPOSED ADVERSARIAL DEFENSES

Adversarial attacks pose significant threats to intelligent NIDS, and, to date, defenses are still limited. This section reviews existing countermeasures, which can be mainly categorized into Adversarial Training and Architecture Reinforcing. Table 4 summarizes the strengths and limitations of the reviewed defenses from the literature.

6.1 Adversarial Training

Adversarial training [25] is a method used to enhance the robustness of machine learning models by including adversarial examples, inputs deliberately altered with small, often imperceptible changes aimed at misleading the model—during the training process. These examples challenge the model by exposing its vulnerabilities, forcing it to learn from both standard and adversarial inputs. As a result, the model becomes more resilient to adversarial

表3. 對NIDS的對抗攻擊總結

參考文獻	Year	攻擊類型			適用性		限制
		白盒	黑盒	灰	目標模型	資料集	
[105]	2023	✓	✗	✗	MLP	CIC-IDS2017，CIC-IDS2018，CIC-DDoS2019	未考慮特徵間相依性，缺乏現實性
[17]	2020	✓	✗	✗	MLP	CICIDS 2017，TRAbID 2017	未考慮特徵間相依性，缺乏現實性
[117]	2020	✓	✗	✗	DNN	CICIDS 2017	未考慮特徵間相依性 將負值引入某些網路流量特徵的介紹
[62]	2019	✓	✗	✗	DNN	CICIDS 2017	泛化能力與現實性不足
[11]	2021	✓	✗	✗	RF, J48	電力系統	處理特徵相關性方面的細節不足 由於預訓練模型，泛化能力有限 特徵修改的平衡難度
[7]	2021	✓	✗	✗	SVM、RF、DT、NB、MLP、KNN、LR、GB、Bagging、LDA、QDA	NSL-KDD，UNSW-NB15	對實時攻擊的影響
[153]	2022	✗	✓	✗	C-LSTM、MLP、 CNN	CSE-CICIDS 2018	沒有關於特徵選擇和相關性的解釋 攻擊者查詢過程的資訊有限
[119]	2019	✗	✓	✗	DNN	KDDcup99，CICIDS2017	傳統距離度量方法的效能有限 關於馬哈拉諾比斯距離參數最佳設定的細節不足 關於攻擊者如何收集目標模型資訊的細節缺失
[146]	2019	✗	✓	✗	CNN & DT	CTU13	有限動作空間探索 無法直接利用特徵相關性
[44]	2023	✗	✓	✗	DNN	Ad-Hoc	微妙的、可偵測的干擾 缺乏攻擊者如何收集資訊的說明
[161]	2022	✓	✓	✗	DNN	NSL-KDD，CICIDS 2017	使用有限的數據對代理模型，降低攻擊成功率
[86]	2022	✗	✓	✗	SVM、MLP、KNN、RF、LR	NSL-KDD	過時數據，缺乏現實性
[129]	2022	✗	✓	✗	LSTM	CIC-IDS2017，Kitsune	複雜實時設置（靈活性較低）
[144]	2022	✗	✗	✓	MLP、RF	CIC-IDS2017，IoT-23	若A2PM使用的訓練資料不足，則生成的攻擊將會不切實際且被偵測
[78]	2022	✓	✓	✓	AdaBoost、Bagging、Gradient Boosting、LR、DT、RF、SVC	NSL-KDD，UNSW-NB 15	未考慮特徵間的相依性，缺乏現實性
[130]	2022	✓	✗	✓	*ML (未指定)*	NSL-KDD，UNSW-NB 15	過時數據，缺乏現實性

- 攻擊範圍：對抗性干擾是在哪個空間中產生的？在問題空間（網絡交通) 或在特徵空間 (學習的模型特徵)？

在強健型NIDS領域的進一步研究應謹慎考慮目標網絡的具體類型，以及敵方可預期的訪問級別和知識。所考慮的威脅模型應設計為反映這些考慮因素，而分析中使用的對抗攻擊必須與威脅模型中概述的訪問級別一致。

6 對所提出對抗防禦的分析

對抗攻擊對智能型NIDS構成顯著威脅，而至今防禦手段仍顯有限。本節檢視現有的對抗措施，這些措施主要可分為對抗訓練和架構加強。表4總結了文獻中審視的防禦措施的優點和限制。

6.1 對抗訓練

對抗訓練 [25] 是一種方法，透過包含對抗樣本來增強機器學習模型的穩固性，這些樣本是在訓練過程中故意進行微小且通常難以察覺的改變，旨在欺騙模型——。這些樣本會挑戰模型的弱點，迫使它從標準和對抗輸入中學習。因此，模型對對抗

attacks, reducing the likelihood of being fooled by manipulated data and improving its overall generalization and performance in more complex, real-world scenarios.

Alahmed et al. [5] propose an adversarial defense method that utilizes Random Forest classifiers for NIDS. Their approach includes the training of a Generative Adversarial Network (GAN) to generate realistic adversarial network traffic examples. The NIDS model is then trained in two phases: first on real traffic data, and subsequently on a combined dataset that includes both the adversarial examples produced by the GAN and the original data. This two-step training process bolsters the model's defenses against adversarial perturbations. The approach is further enhanced through Principal Component Analysis (PCA) for feature selection, and the model's resistance is evaluated using a black-box attack technique. Despite its strengths, this approach suffers from various drawbacks and limitations. Firstly, the reliance on GANs to produce synthetic network data is costly and inefficient. GANs are shown to be tricky to train and stabilize. Secondly, and more crucially, the GAN-generated network data are not validated as real network traffic flow data. The fact that GANs are able to generate data from a given distribution does not mean that the generated data represents real-world traffic flows. Similarly, Debicha et al [45] aim at strengthening a DL-based NIDS through the use of adversarial training. To generate the adversarial examples, they rely on three off-the-shelf adversarial generation techniques, namely the FGSM, PGD, and BIM. Similarly to the previous approach, the proposed technique does not consider the constraints required for the generated adversarial network flow to be realistic. Additionally, it shares the same limitation of having limited practical application. Venturi et al. [143] address network intrusion detection by monitoring network traffic through a graph representation, including both topology and individual connections. The approach uses a Graph Neural Network (GNN) within an AE to condense graph complexities into embeddings that capture important components of each flow and their interactions, acting as a feature extractor. To improve resilience against data perturbations, they rely on adversarial training as well, exposing the AE to both legitimate and malicious traffic to learn robust embeddings. These embeddings, containing network architecture and flow details, are then processed by an RF classifier for normal and abnormal traffic classification. A key limitation to this procedure is its high computational cost, especially in resource-constrained environments; therefore, optimizing this process for greater efficiency represents a viable direction for building adversarially robust NIDS. Another approach, the EIFDAA framework [85], provides active defense through adversarial training based on three steps. First, a simulation of attacks in which adversarial examples are generated using BIM, PGD, FGSM, and DeepFool. These attacks modify legitimate data to create malicious samples that mimic typical traffic patterns and evade IDS detection. The second step is adversarial evaluation, which requires including original attack data that represents real attacks together with simulated attacks into the IDS being evaluated. A significant decrease in attack detection indicates an IDS vulnerability. Finally, adversarial training allows the IDS to learn attacker techniques and fortify its defenses against future attempts to exploit similar vulnerabilities by retraining it using a combination of adversarial examples and the original training data. EIFDAA offers a comprehensive evaluation and defensive method. However, its performance depends on the accuracy of the evaluation data and the quality of attack simulations.

Recently, Debicha et al. [44], aimed at crafting a filter-like system to prevent adversarial examples from reaching and fooling the NIDS. This detector acts as a pre-filter using three techniques: adversarial training, adversarial detection, and ensemble methods. Three MLPs are trained in parallel; each one focuses on a specific set of network traffic features based on the possibility of manipulating them by the attacker (modifiable or unmodifiable). The training data is composed of two classes: adversarial and clean. During detection, each MLP monitors an incoming data point. Each prediction is then weighted based on its training accuracy. The weighted predictions from all three MLPs are integrated using a combination of contextual discounting and Bayesian fusion, leading to a final classification of each data point as either *adversarial* or *clean*. This classification determines whether the data point is permitted to pass through to the NIDS. While effective, running multiple models can be computationally expensive, and accurately classifying features may be difficult due to manipulation by attackers.

攻擊變得更加強韌，降低被操縱數據騙局的機率，並提高其在更複雜、真實世界情境中的整體泛化能力和性能。

Alahmed 等人 [5] 提出一種利用隨機森林分類器進行網路入侵偵測系統 (NIDS) 的對抗性防禦方法。他們的方法包括訓練一個生成對抗網絡 (GAN) 以生成真實的對抗性網路流量範例。接下來，NIDS 模型分兩個階段進行訓練：首先在真實流量數據上，然後在包含 GAN 生成的對抗範例和原始數據的組合數據集上。這兩個步驟的訓練過程增強了模型對對抗干擾的防禦能力。這種方法通過主成分分析 (PCA) 進行特徵選擇得到進一步的增強，並使用黑盒攻擊技術評估模型的抗性。儘管有其優點，但這種方法存在各種缺點和限制。首先，依賴 GAN 生成合成網路數據既昂貴又低效。證明 GANs 很難訓練和穩定。其次，更重要的是，GAN 生成的網路數據未被驗證為真實的網路流量數據。GANs 能夠從給定分佈生成數據並不意味著生成的數據代表真實世界的流量流。類似地，Debicha 等人 [45] 旨在通過使用對抗性訓練來增強基於深度學習的 NIDS。為了生成對抗範例，他們依賴三種即裝即用的對抗生成技術，即 FGSM、PGD 和 BIM。與先前的方法類似，所提出的技術不考慮生成對抗性網路流量為真實所需的約束。此外，它與先前的方法一樣存在實際應用有限的相同限制。Venturi 等人 [143] 通過監控包含拓撲和個別連接的圖表示來處理網路入侵偵測。該方法在 AE 中使用圖神經網絡 (GNN) 將圖複雜性壓縮為嵌入，這些嵌入捕捉了每個流及其交互中的重要組成部分，作為特徵提取器。為了提高對數據干擾的韌性，他們也依賴對抗性訓練，將 AE 暴露於合法和惡意流量中，以學習堅固的嵌入。這些包含網路架構和流細節的嵌入，然後由 RF 分類器處理，用於正常和異常流量的分類。此程序的一個關鍵限制是其高計算成本，尤其是在資源受限的環境中；因此，優化此過程以提高效率代表了建立對抗性強健 NIDS 的可行方向。另一種方法，EIFDAA 框架 [85], 通過基於三個步驟的對抗性訓練提供主動防禦。首先，模擬攻擊，其中使用 BIM、PGD、FGSM 和 DeepFool 生成對抗範例。這些攻擊修改合法數據以創建模擬典型流量模式並逃避免偵測的惡意樣本。第二步是對抗評估，這需要將代表真實攻擊的原始攻擊數據與模擬攻擊一起包含在要評估的 IDS 中。攻擊檢測顯著下降表明 IDS 存在漏洞。最後，對抗性訓練允許 IDS 學習攻擊者技術並通過使用對抗範例和原始訓練數據的組合重新訓練來加強其防禦，以防止未來嘗試利用類似漏洞。EIFDAA 提供了一種全面的評估和防禦方法。然而，其性能取決於評估數據的準確性和攻擊模擬的質量。

最近，Debicha 等人 [44], 旨在打造一個過濾器般的系統，以防止對抗性範例抵達並騙取 NIDS。這個偵測器作為一個預過濾器，使用三種技術：對抗性訓練、對抗性偵測和集成方法。三個 MLP 並行訓練；每個 MLP 都專注於一組特定的網路流量特徵，這些特徵基於攻擊者（可修改或不可修改）操控的可能性。訓練數據由兩個類別組成：對抗性和乾淨。在偵測期間，每個 MLP 監控一個進來的數據點。然後，每個預測都根據其訓練準確性進行加權。所有三個 MLP 的加權預測使用上下文折現和貝葉斯融合的組合進行整合，導致每個數據點的最終分類為對抗性或乾淨。這個分類決定數據點是否被允許通過到 NIDS。雖然有效，但運行多個模型可能計算成本高昂，並且由於攻擊者的操控，準確分類特徵可能很困難。

TAKEAWAY. Adversarial training has become one of the most widely used technique for enhancing the robustness of machine learning models, particularly DNNs. This is due to its ease of use, its flexibility, and its demonstrated effectiveness in improving model resilience. However, several considerations must be taken into account when implementing adversarial training within ML pipelines. Firstly, there is a risk of overfitting. Adversarial training alters the decision boundary learned by the model, with the goal of making it more robust to malicious outliers (adversarial examples). However, incorporating an excessive number of adversarial examples or allowing overly large perturbations poses the risk of excessively altering the decision boundary, which can lead to misclassifications and diminished generalization ability. Secondly, and most importantly, the adversarial sample generation techniques must be carefully selected and tuned to ensure that the generated samples are meaningful representations of actual network traffic, relevant to the task of NIDS. Particularly, to maximize the robustness of the model, it is crucial to consider feature constraints and interdependencies during adversarial example generation.

6.2 Architecture Reinforcing

A novel area of research in defending against adversarial attacks is architecture reinforcing [59, 67]. This field focuses on developing neural network architectures and designing models that are inherently robust to adversarial attacks by design.

Jiang et al. [77] rely on multi-model and task modelling to achieve adversarial robustness. They analyze network data and divide features into three groups: time-related, packet-related, and state-related, which are less likely to be altered by attackers. Each group is processed by a separate DL model, and the predictions are combined to decide if a flow is malicious. This approach shows good adversarial robustness, while maintaining adequate detection rates in the absence of adversarial attacks. However, its effectiveness depends on accurately predicting which features attackers will target, which can reduce its reliability when adversaries alter their attack.

Jmila et al. [78] assessed the robustness of Gaussian data augmentation against adversarial attacks by introducing random Gaussian noise into the training data. This method aims to expand the dataset and limit the ability of attackers to craft targeted adversarial examples, making the IDS less vulnerable to feature manipulation. The authors compared IDS classifiers trained on original versus augmented data, evaluating them on both normal and adversarial test sets. They measured the differences in accuracy and False Negative Rate to assess the model's performance. While this approach enhances IDS resilience, its effectiveness may be constrained by the inability to fully simulate real-world adversarial manipulations.

Zhang et al. [152] propose another method that incorporates three strategies to defend against different types of black-box adversarial attacks, namely ensemble voting, ensemble adversarial training and adversarial sample detection, with each strategy having its limitations. For instance, combining multiple deep learning models can increase complexity and heighten the risk of misclassification. Adversarial training, while useful, relies heavily on the attacker's knowledge and risks overfitting. Query detection requires careful threshold tuning and remains vulnerable to attacks that can bypass the initial detection phase. Additionally, these methods raise computational costs and may struggle to keep pace with rapidly evolving threats.

In contrast to traditional models that explicitly classify attack types, a two-layer hierarchical defense system was proposed by Mccarthy et al. [98]. It uses a coarse layer to group similar attacks (e.g., various DoS attacks) and a fine layer to classify specific types (e.g., Slowloris). This makes it challenging for attackers to inject perturbations that trick both layers. Despite the high computational cost of this approach and the need to define the attack group hierarchy, it results in a robust model that prevents adversarial examples and maintains high classification accuracy in regular traffic.

Recently, Mutation-based detection has proven effective for adversary defense [63] as have approaches that rely on observing adversarial manipulations occurring near the decision boundary [132]. Wang et al. [145] identify

重點。對抗性訓練已成為增強機器學習模型（尤其是 DNN）魯棒性最受歡迎的技術之一。這是因為其易用性、靈活性以及在提高模型韌性方面所證明的有效性。然而，在 ML 管道中實施對抗性訓練時，必須考慮幾個因素。首先，存在過度擬合的風險。對抗性訓練改變模型學習的決策邊界，目的是使其對惡意異常（對抗性範例）更為穩固。然而，包含過多的對抗性範例或允許過大的擾動會導致決策邊界過度改變，這可能導致錯誤分類並降低泛化能力。其次，最重要的是，對抗性範例生成技術必須謹慎選擇和調整，以確保生成的範例是實際網路流量的有意義表示，與 NIDS 的任務相關。特別是，為了最大化模型的穩固性，在對抗性範例生成期間考慮特徵約束和相互依存性至關重要。

6.2 強化架構

在防禦對抗性攻擊方面，一個新的研究領域是架構強化 [59, 67]。這個領域專注於開發神經網路架構和設計天生對抗性攻擊具有韌性的模型。

Jiang 等人 [77] 依賴多模型和任務建模來實現對抗性韌性。他們分析網路數據，將特徵分為三組：與時間相關的、與封包相關的和與狀態相關的，這些特徵不太可能被攻擊者篡改。每一組都由一個獨立的DL模型處理，並將預測結果組合起來判斷流量是否惡意。這種方法展現了良好的對抗性韌性，同時在沒有對抗性攻擊的情況下仍能保持足夠的偵測率。然而，它的有效性取決於準確預測攻擊者將目標鎖定的特徵，當攻擊者改變攻擊方式時，這會降低其可靠性。

Jmila 等人 [78] 通過將隨機高斯雜訊引入訓練數據，評估了高斯數據增強對對抗性攻擊的韌性。這種方法旨在擴展數據集並限制攻擊者製造針對性對抗性範例的能力，從而降低IDS對特徵操縱的易受性。作者比較了在原始數據和增強數據上訓練的IDS分類器，並在正常和對抗性測試集上評估它們的性能。他們測量了準確率和假陰性率的差異來評估模型的性能。雖然這種方法增強了IDS的韌性，但其有效性可能受到無法完全模擬真實世界對抗性操縱的限制。

張等人 [152] 提出一種新方法，整合三種策略來防禦不同類型的黑盒對抗攻擊，即集成投票、集成對抗訓練和對抗樣本偵測，每種策略都有其限制。例如，結合多個深度學習模型會增加複雜性並加劇錯分風險。對抗訓練雖然有用，但高度依賴攻擊者的知識，且存在過度擬合的風險。查詢偵測需要仔細調整閾值，且仍易受能夠繞過初始偵測階段的攻擊。此外，這些方法會增加計算成本，且可能無法跟上快速演變的威脅。

與傳統的明確分類攻擊類型的模型不同，麥卡蒂等人 [98] 提出一種兩層階層式防禦系統。它使用粗層來分組類似的攻擊（例如各種 DoS 攻擊），並使用細層來分類特定類型（例如 Slowloris）。這使得攻擊者難以注入能騙過兩層的干擾。儘管這種方法的計算成本很高，且需要定義攻擊分組階層，但它能產生一個穩固的模型，防止對抗樣本並在正常流量中維持高分類準確率。

最近，基於變異的偵測已被證明對威脅防禦 [63] 有效，而依賴觀察決策邊界附近發生的威脅操縱的方法 [132] 也同樣有效。王等人 [145] 識別

Table 4. Comprehensive summary: Analytical study of adversarial defense strategies to NIDS

Reference	Year	Defense Method	Strengths	Limitation
[5]	2022	**Adversarial Training**	High detection rate of FGSM perturbations, while maintining the original detection level	High training cost and limited generalizability
[45]	2021		Detection for some specific adversarial attacks	Reliance on outdated data and minimal adversarial training
[143]	2023		Strong against specific adversarial attacks	Computational cost, limited generalizability, and limited adversarial training
[85]	2023		Strong against some specific adversarial attack	Limited generalizability and reliance on accurate data
[44]	2023		High detection performance	Computationally expensive, vulnerable to evolving attacks
[77]	2022	**Architecture Reinforcing**	Improved attack detection through multi-modal analysis	Limited robustness to adaptive and evolving attacks
[78]	2022		Simplicity and improved generalizability	Loss of information, restricted to specific attacks
[152]	2020		High detection performance	Computationally expensive and limited adaptability
[98]	2023		Strong robustness to evasion	Computationally expensive
[145]	2022		High true positive rate with low false positives	Computationally expensive Its effectiveness depends on the accuracy of the target model Not adaptive to more sophisticated attacks

adversarial perturbations by detecting inconsistencies between the NIDS's classification and natural data patterns (manifolds). These manifolds represent the original data structure and decision boundary, making it hard to detect perturbations as they share characteristics with both malicious and benign traffic. The effectiveness of this approach relies on the accuracy of the IDS model, leaving it vulnerable to exploitation if the model has any weaknesses. Moreover, this dual approach, involving manifold analysis and decision boundary detection, is more computationally expensive compared to previous detection techniques.

TAKEAWAY. Rather than focusing on specific attack patterns, the architecture reinforcing of NIDS aims to develop more generalized and adaptable defenses. This broader approach enhances the system's resilience to a wide range of emerging threats, reducing the need for frequent updates to address new attack types. Additionally, this strategy simplifies model maintenance and reduces the need for constant retraining. However, it requires careful consideration of trade-offs between security, accuracy, and complexity. Striking the right balance is essential to ensure the system effectively counters evolving threats without becoming excessively resource-intensive or cumbersome. Furthermore, differently from adversarial training, robust architectures need to be designed specifically for a target task and setting, limiting their versatility.

6.3 Discussion

Despite the rapid advancements in adversarial machine learning, there remains a substantial gap in the development and implementation of effective defenses for NIDS. Current research predominantly emphasizes the analysis of adversarial attacks, with comparatively less focus on devising practical and resilient defense mechanisms. Adversarial training has emerged as a promising approach for enhancing the robustness of NIDS against adversarial examples. However, its effectiveness heavily relies on the quality and diversity of the training data, making the process resource-intensive. Furthermore, excessive use of adversarial training may lead to overfitting and reduced generalization capabilites. Alternatively, approaches leveraging robust architectures such as ensembles methods or robust loss functions have been proposed to improve resistance to adversarial manipulation. Nevertheless, these methods are often inadequate in addressing the continuous evolution of adversarial attacks. Hybrid strategies integrating adversarial training with robust architectures may lead to more robust NIDS solutions. However, the current lack of understanding regarding the root causes of adversarial examples, coupled with the limited research on robustness in structured data domains, complicates the identification of potential solutions. Given the dynamic and evolving nature of adversarial threats and the peculiar characteristics of structured data, further research is required to identify promising defense mechanisms to improve the robustness of NIDS.

表4。綜合總結：針對NIDS的對抗防禦策略分析研究

參考年分		防禦方法	優點	限制
[5]	2022	**對抗訓練**	對FGSM干擾的高偵測率，同時維持原本的偵測水平	訓練成本高且泛化能力有限
[45]	2021		針對某些特定對抗攻擊的偵測	依賴過時數據和極少的對抗訓練
[143]	2023		對特定對抗性攻擊有強韌性	計算成本高、泛化能力有限，以及對抗性訓練受限
[85]	2023		對某些特定對抗性攻擊有強韌性	泛化能力有限，且依賴準確數據
[44]	2023		檢測性能高	計算成本高，易受演變中的攻擊影響
[77]	2022	**建築強化**	透過多模態分析提升攻擊偵測	對於適應性和演變中的攻擊，魯棒性有限
[78]	2022		簡易性與提升的泛化能力	資訊損失，限制於特定攻擊
[152]	2020		高偵測性能	計算成本高且適應性有限
[98]	2023		對迴避攻擊有強韌性	計算成本高
[145]	2022		真陽性率高且偽陰性率低	計算成本高 其有效性取決於目標模型的準確性 不適應更複雜的攻擊

威脅干擾，方法是偵測網路入侵偵測系統(NIDS)的分類與自然數據模式（流形）之間的不一致。這些流形代表原始數據結構和決策邊界，使得偵測干擾變得困難，因為它們與惡意和良性流量都具有相似特徵。此方法的效能取決於IDS模型的準確性，如果模型有任何弱點，就會容易受到利用。此外，這種涉及流形分析和決策邊界偵測的雙重方法，相較於先前的偵測技術，在計算成本上更為昂貴。

TAKEAWAY. 與其專注於特定的攻擊模式，加強NIDS架構的目標是發展更通用且適應性強的防禦措施。這種更廣泛的方法增強了系統對各種新興威脅的韌性，減少了為應對新的攻擊類型而頻繁更新的需求。此外，這種策略簡化了模型維護，並減少了持續重新訓練的需求。然而，它需要仔細考慮安全、準確性和複雜性之間的權衡。達到正確的平衡至關重要，以確保系統能有效對抗演變中的威脅，而不會變得過度資源密集或麻煩。此外，與對抗性訓練不同，穩固的架構需要針對特定的任務和環境進行專門設計，這限制了它們的靈活性。

6.3 討論

儘管對抗式機器學習快速發展，但在NIDS（網路入侵偵測系統）的開發與實施有效防禦方面，仍存在顯著差距。當前研究主要強調對抗式攻擊的分析，相較之下對於設計實用且強韌的防禦機制則較少關注。對抗式訓練作為一種增強NIDS對抗對抗式樣本強韌性的前景光明的方法已經出現。然而，其有效性高度依賴訓練數據的品質與多樣性，導致過程資源密集。此外，過度使用對抗式訓練可能導致過度擬合與泛化能力的降低。另方面，利用強韌架構的方法，例如集成方法或強韌損失函數，已被提出以提升對抗式操縱的抵抗力。然而，這些方法往往無法應對對抗式攻擊的持續演變。整合對抗式訓練與強韌架構的混合策略或許能帶來更強韌的NIDS解決方案。然而，當前對於對抗式樣本根本原因的缺乏理解，加上對結構化數據領域強韌性的研究有限，複雜化了潛在解決方案的識別。考慮到對抗式威脅的動態與演變性質，以及結構化數據的特殊特性，需要進一步研究以識別有前景的防禦機制，以提升NIDS的強韌性。

7 LESSONS LEARNED, NEW INSIGHTS AND GUIDELINES FOR FUTURE RESEARCH

This section explores lessons learned and potential future research directions, providing insights and guidelines to improve future research in the area.

7.1 Inter-dependency of Structured Features

As discussed in Section 5, existing studies often fail to consider the potential impact of feature constraints and correlations when designing adversarial attacks against NIDS. This leads to a gap between theoretical analysis and recommendations, and their practical feasibility. Key characteristics that require consideration include:

- **Syntactic and Semantic Validity:** The attack might be invalid if packet headers or content are modified without considering the context of the network communication. Network protocols have well-defined syntactic and semantic rules that dictate the structure and behavior of traffic. Arbitrary modifications to packet data may lead to violations of these protocols.
- **Time-based dependencies:** Network traffic often presents time-sensitive patterns. If adversarial perturbations incorrectly alter these temporal dependencies (e.g., sending an acknowledgment before receiving the associated data packet), it will lead to protocol violations and lead to failure in real-world settings.
- **Dependent features:** Modifications to certain attributes, such as packet size or frequency, lead to changes in other features, such as total number of bytes or communication duration. Adversarial attacks and defenses must consider these dependencies to be effective in real-world settings.

Practical adversarial attacks against NIDS must consider these characteristics to ensure realistic analysis and results that can translate into real-world settings. Neglecting these factors may lead to findings that are disconnected from real-world scenarios.

7.2 Lack of Defense Strategies against Adversarial Attacks

This survey highlights the lack of robust defenses capable of addressing various types of adversarial attacks. Multiple factors contribute to this shortcoming, posing a significant and persistent challenge for ML models:

- **Lack of explainability:** DL models often lack interpretability and transparency due to their complex decision-making processes and architectures, making it challenging to understand their internal functions [60]. This hinders the development of effective defenses against adversarial attacks, as the models' weaknesses are not fully understood.
- **Dynamic nature of adversarial examples:** Adversarial attacks are constantly evolving as attackers develop new strategies to exploit models' vulnerabilities. This evolution is driven by the open nature of research, and the availability of large model architectures and datasets, which enable attackers to refine their techniques [133]. Consequently, static defenses face a significant challenge in keeping up with the rapidly changing landscape of these attacks.
- **Generalization:** Existing defenses against adversarial examples are developed with specific attack scenarios and datasets, making them difficult to apply to other contexts and real-world settings. Therefore, these countermeasures may not generalize well and might fail against new attack variations on different datasets.
- **Transferability:** Adversarial manipulations created for one model can often be successfully transferred to other models, a challenge known as *transferability* [147]. Hence, an effective defense for one model may be ineffective for another with similar vulnerabilities against new attack iterations within different datasets.

Intelligent models remain susceptible to adversarial attacks, which prevents their safe deployment in cybersecurity. This weakness stems from both the complexity of these attacks and the incomplete understanding of how they exploit model flaws. To ensure the safe use of machine learning, further research is needed on defense strategies and the explainability of adversarial attack behaviors.

7 個學習心得、新見解與未來研究指導原則

本節探討經驗教訓與潛在的未來研究方向，提供洞察與指導原則，以提升該領域未來研究的品質。

7.1 資料結構的相互依存性

如第 5 節所述，現有研究在設計針對 NIDS 的對抗性攻擊時，往往未能考慮特徵限制和相關性的潛在影響。這導致了理論分析與建議之間的差距，以及其實際可行性。需要考慮的關鍵特徵包括：

- 語法和語義有效性：如果未考慮網絡通信的語境而修改封包標頭或內容，攻擊可能無效。網絡協議具有明確定義的
  語法和語義規則，它們規定了流量結構和行為。對封包數據的隨意修改可能導致違反這些協議。
- 時間相依性：網路流量常呈現時間敏感的模式。若敵對的干擾錯誤地改變這些時間相依性（例如，在接收相關數據包之前就發送確認訊息），將會導致協議違規，並導致在實際應用場景中失敗。
- 相依特徵：對某些屬性（例如包大小或頻率）的修改，會導致其他特徵（例如總字數或通訊持續時間）的變化。敵對攻擊和防禦必須考慮這些相依性，才能在實際應用場景中發揮效果。

針對NIDS的實際敵對攻擊必須考慮這些特性，以確保實際的分析和結果能夠轉化為實際應用場景。忽略這些因素可能導致發現與實際應用場景脫節。

7.2 缺乏對敵對攻擊的防禦策略

本調查突顯了缺乏能夠應對各種類型敵對攻擊的穩固防禦措施。多種因素導致了這一短處，對機器學習模型構成了嚴重且持續的挑戰：

- 缺乏解釋性：DL模型因其複雜的決策過程和架構，通常缺乏可解釋性和透明度，導致難以理解其內部功能 [60]。這阻礙了對抗性攻擊的有效防禦開發，因為模型的弱點並未完全被理解。
- 對抗性範例的動態性：隨著攻擊者開發新的策略來利用模型的弱點，對抗性攻擊不斷演變。這種演變是由於研究的開放性，以及大型模型架構和數據集的可用性，這些使得攻擊者能夠精煉其技術 [133]。因此，靜態防禦面臨著在追趕這些攻擊快速變化的環境時的重大挑戰。
- 泛化能力：現有的對抗性範例防禦措施是針對特定的攻擊場景和數據集開發的，使其難以應用於其他背景和實際環境。因此，這些對策可能泛化能力不佳，並可能在不同的數據集上對新的攻擊變體失效。
- 可遷移性：針對一個模型創建的對抗性操縱，通常可以成功地遷移到其他模型，這是一個所謂的可遷移性挑戰 [147]。因此，針對一個模型的有效防禦可能對另一個具有相似弱點的模型在新的數據集上的攻擊迭代中無效。

智慧型模型仍然容易受到對抗性攻擊，這些攻擊阻礙了它們在網路安全的安穩部署。這種弱點源於這些攻擊的複雜性，以及對於它們如何利用模型缺點的不完整理解。為了確保機器學習的安全使用，需要進一步研究防禦策略和對抗性攻擊行為的可解釋性。

7.3 Lack of Real-World Relevance

The analytical study conducted in Section 5 identifies a critical gap in the literature: the lack of real-world applicability. This gap arises from several factors:

- **Dataset limitations:** Some datasets are outdated and/or fail to capture the complexity and diversity of real-world network traffic, which undermines the credibility of experimental findings.
- **Simplified attack techniques:** Most existing studies consider complete control over datasets and full knowledge of the target NIDS, which does not reflect real-world conditions.
- **Evolving Attack Strategies:** Researchers frequently focus on established adversarial attack methods, overlooking the adaptability and creativity of real-world attackers, who might exploit previously undiscovered vulnerabilities, compromising the effectiveness of current defense strategies.

By addressing these limitations, future research can provide more practical insights to enhance the development of robust defenses against emerging cyber threats.

7.4 Placement of Adversarial Examples

Assuming data can be represented as collections of manifolds [132], many studies explored the vulnerabilities of adversarial examples. They characterized adversarial subspaces and examined the relationship between adversarial perturbations and manifolds. These investigations can be categorized into the following groups:

- **Leave-the-manifold attacks:** They create perturbations that force input data outside the manifold (or learned data distribution). Researchers (e.g., [53, 76, 126, 128, 136]) examined this concept, noting that these modifications are more intense and introduce features not in the training dataset. They are larger and more visible, exploiting the model's weaknesses in handling unfamiliar or out-of-distribution data.
- **On-manifold attacks:** Another possible position of adversarial attacks is observing them on-manifold [132], as it is illustrated in Figure 6. This concept states that training data and adversarial samples belong to the same distribution or manifold. In other words, these attacks stay within the feature space that the model is familiar with from training. Compared to Leave-the-manifold attacks, these perturbations are usually subtle and do not visibly alter the input data.
- **Low-dimensional adversarial subspaces:** This option refers to adversarial examples existing in a lower-dimensional subspace of the feature space [90]. By exploiting this structure, attackers can find perturbations that significantly affect the model's output with minimal input modifications.

Insight. These findings have been extensively explored in the image domain, where the input space is mostly unconstrained and there are soft interdependencies between features. Structured data such as network traffic, however, is typically characterized by hard feature interdependencies and constraints that can significantly affect the space of allowable adversarial examples and their placement. This research gap presents an opportunity to explore a better characterization of adversarial examples in the context of structured data. Current defenses against adversarial attacks used in NIDS are typically based on heuristic approaches that do not necessarily take into consideration the localization of adversarial examples in the domain space. Gaining a deeper understanding of adversarial perturbations and the relationship between feature interdependencies and the positioning of adversarial examples within the network data space can lead to significant improvement in NIDS robustness.

7.5 Good Practices

As the volume and complexity of data that must be processed per unit of time continue to increase, systems such as NIDS must utilize techniques like ML for rapid and efficient detection. However, several factors need to be taken into account before implementing ML-based approaches for network intrusion detection:

7.3 缺乏實際應用性

第5節進行的分析研究識別出文獻中的一個關鍵差距：缺乏實際應用性。這個差距源於多個因素：

- 數據集限制：部分數據集已過時且/或無法捕捉實際網路流量複雜性和多樣性，這削弱了實驗結果的可信度。
- 簡化的攻擊技術：大多數現有研究考慮對數據集的完全控制以及對目標NIDS的全面了解，這並未反映實際情況。
- 演進中的攻擊策略：研究人員經常專注於既有的對抗攻擊方法，忽略了實際攻擊者的適應性和創造力，而實際攻擊者可能利用先前未發現的漏洞，損害當前防禦策略的有效性。

透過解決這些限制，未來研究可以提供更具實際性的洞見，以加強對新興網路威脅的堅固防禦開發。

7.4 對抗範例的放置

假設資料可以表示為流形集合 [132]，許多研究探討了對抗範例的弱點。他們描述了對抗子空間，並檢視了對抗干擾與流形之間的關係。這些調查可以分類為以下幾類：

- 離開流形攻擊：它們產生干擾，迫使輸入資料離開流形（或學習的資料分佈）。研究人員（例如，[53, 76, 126, 128, 136]）檢查了這個概念，指出這些修改更強烈，並引入了訓練資料集中不存在的特徵。它們更大、更明顯，利用了模型在處理不熟悉或非分佈資料時的弱點。
- 在流形上攻擊：另一種可能的對抗攻擊位置是觀察在流形上 [132]，如圖6所示。這個概念認為訓練資料和對抗樣本屬於相同的分佈或流形。換句話說，這些攻擊始終停留在模型從訓練中熟悉的特徵空間內。與離開流形攻擊相比，這些干擾通常很微細，不會明顯改變輸入資料。
- 低維對抗子空間：此選項指存在於特徵空間低維子空間中的對抗樣本 [90]。透過利用此結構，攻擊者可以找到對模型輸出產生顯著影響的擾動，而輸入修改最小。

洞見。這些發現已在影像領域得到廣泛探索，其中輸入空間大多不受限制，且特徵之間存在軟性相關性。然而，結構化數據（如網路流量）通常具有嚴格的特徵相關性與限制，這些會顯著影響允許的對抗樣本空間及其位置。這個研究缺口為在結構化數據背景下更好地描述對抗樣本提供了機會。目前用於NIDS的對抗攻擊防禦通常基於啟發式方法，不一定考慮對抗樣本在領域空間中的定位。更深入地理解對抗擾動以及特徵相關性與網路數據空間中對抗樣本定位之間的關係，有助於顯著提升NIDS的穩健性。

7.5 優良實踐

隨著每單位時間內必須處理的數據體量和複雜性持續增加，NIDS等系統必須利用機器學習等技術進行快速高效的偵測。然而，在實施基於機器學習的網路入侵偵測方法之前，需要考慮以下幾個因素：

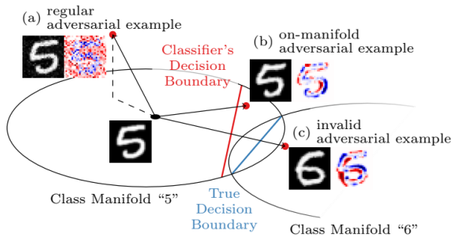


Fig. 6. Comparative illustration of standard, on-manifold *(constrained)* adversarial examples vs. unconstrained ones [132].

- **Adversarial training:** Incorporating adversarial perturbations during the training process of a NIDS inherently increases robustness and resilience against evasion attempts during inference. However the adversarial training should be carefully done by using plausible adversarial data and not just rely on simplistic adversarial generation techniques that would fool the machine learning model but do not represent valid network data. This would allow the model to actually become robust towards attacks that might actually happen, not just overfit on non-plausible adversarial data.
- **Ensemble learning and fusion:** In addition to adversarial training, ensemble learning approaches have proven effective against adversarial attacks. Combining multiple diverse architectures, each trained differently, allows them to learn slightly different decision boundaries, improving overall performance and robustness. The independent errors of these models contribute to better performance and increased overall adversarial robustness since not all models are affected similarly by the same adversarial sample. Robustness can be further tuned through different model fusion methods (e.g., voting methods, averaging, and meta-learning), leveraging the strengths of individual models. For ensemble-based NIDS, carefully studying the resilience of individual models to several adversarial attacks can help tailor model fusion strategies to enhance adversarial robustness, making NIDS less vulnerable to attacks.
- **Robust feature engineering:** Incorporating high-level semantic features vs. low-level or raw data in NIDS. Incorporating high-level semantic features rather than low-level or raw data in NIDS generally provides a more robust and resilient foundation for machine learning models. Focusing on semantic understanding and abstraction improves model stability, generalization, and resistance to adversarial attacks, increasing reliability and effectiveness in real-world applications.
- **Input preprocessing:** Input preprocessing plays a crucial role in enhancing the robustness of machine learning models against adversarial attacks. By applying techniques such as data normalization, feature scaling, noise filtering, and input transformation, preprocessing can reduce the sensitivity of a model to small, malicious perturbations that adversaries introduce to manipulate predictions. These steps help the model focus on the essential patterns within the data rather than being misled by subtle alterations designed to exploit weaknesses. Effective input preprocessing can act as a first line of defense, mitigating the impact of adversarial examples and contributing to a more resilient, reliable system.
- **Anomaly detection:** Anomaly detection can significantly enhance the adversarial robustness of a machine learning model by identifying deviations from expected or normal behavior, thereby flagging potentially malicious inputs. Since adversarial attacks often involve introducing subtle, crafted perturbations designed to mislead the model, anomaly detection techniques can act as a safeguard by detecting these out-of-distribution inputs that do not align with typical data patterns. Regarding ML-based NIDS, envisioning

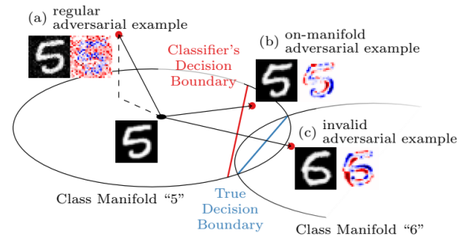


圖 6. 標準、單流形（受限）對抗樣本與非受限樣本的比較說明 [132].

• 對抗訓練：在 NIDS 的訓練過程中引入對抗干擾，可以顯著提高其在推理過程中對迴避嘗試的強韌性和抗性。然而，對抗訓練應謹慎進行，需使用合理的對抗數據，而不僅僅依賴簡單的對抗生成技術，這些技術可能騙過機器學習模型，但並不能代表有效的網絡數據。這樣可以讓模型實際上對可能真正發生的攻擊變得強韌，而不僅僅是過度適應非合理的對抗數據。

• 集成學習與融合：除了對抗訓練，集成學習方法已被證明能有效對抗對抗攻擊。結合多個不同的架構，每個架構訓練方式不同，可以讓它們學習略微不同的決策邊界，從而提高整體性能和強韌性。這些模型的獨立錯誤促進了更好的性能和整體對抗強韌性，因為並非所有模型都受到相同對抗樣本的相似影響。通過不同的模型融合方法（例如，投票方法、平均和元學習），可以進一步調整強韌性，利用個別模型的優勢。對於基於集成的 NIDS，仔細研究個別模型對幾種對抗攻擊的強韌性，有助於調整模型融合策略以增強對抗強韌性，使 NIDS 更少易受攻擊。

• 堅固的特徵工程：在網路入侵偵測系統(NIDS)中整合高階語義特徵與低階或原始數據。在NIDS中整合高階語義特徵而非低階或原始數據，通常為機器學習模型提供更堅固且彈性基礎。專注於語義理解與抽象化可提升模型穩定性、泛化能力及對抗威脅攻擊的抵抗力，增加實際應用中的可靠度與有效性。

• 輸入預處理：輸入預處理在增強機器學習模型對抗威脅攻擊的堅固性上扮演關鍵角色。透過應用數據標準化、特徵縮放、雜訊過濾與輸入轉換等技術，預處理可降低模型對攻擊者引入以操縱預測的微小惡意干擾的敏感度。這些步驟有助於模型專注於數據中的核心模式，而非被旨在利用弱點的微妙變動所誤導。有效的輸入預處理可作為第一道防線，減輕威脅樣本影響，有助於建立更堅固可靠的系統。

• 異常偵測：異常偵測可顯著增強機器學習模型的威脅攻擊堅固性，透過識別與預期或正常行為的偏差，標示潛在的惡意輸入。由於威脅攻擊通常涉及引入微妙、精心設計的干擾以誤導模型，異常偵測技術可作為防護措施，透過偵測不符合典型數據模式的非分佈輸入。關於基於機器學習的NIDS，想像

the detection system as a multi-step approach by having an anomaly filter able to detect these suspicious inputs can be a way to augment the overall performance of the system as a whole.

- **Proper (extensive) evaluation:** Proper model evaluation is critical for ensuring the robustness of a machine learning model, particularly in the face of adversarial attacks. When it comes to NIDS, particular care must be given to evaluating and stress testing the model (and the IDS in general) in scenarios pertaining to the threats it is envisioned to withstand. As such the system should be thoroughly evaluated on not just standard network data, which in many cases are outdated, but on timely data that includes also corner cases, out-of-distribution data that fall under the overall threat surface envisioned in the threat model of the system.

8 CONCLUSION AND FUTURE DIRECTIONS

This survey provides a comprehensive and current analysis of adversarial attacks, emphasizing the dynamic landscape of cybersecurity threats, with a specific focus on network intrusion detection systems. It serves as a valuable resource for researchers, organizations, and cybersecurity professionals, enhancing their understanding of this rapidly evolving field. Additionally, the survey details various evasion techniques employed by adversaries to circumvent NIDS, identifies critical gaps in the existing literature, and outlines emerging challenges and insights that can inform future research and development efforts in this field.

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將偵測系統視為多步驟方法，透過異常過濾器能夠偵測這些可疑輸入，可以作為增強整體系統性能的一種方式。

- 正確（廣泛）的評估：正確的模型評估對於確保機器學習模型的穩固性至關重要，尤其是在面對對抗性攻擊的情況下。針對NIDS而言，必須特別小心評估和壓力測試模型（以及整個IDS），在與其預設要抵禦的威脅相關的場景中。因此，系統應該在標準網路數據上進行徹底評估，而標準網路數據在許多情況下已經過時，同時還應該在包含邊緣案例、屬於系統威脅模型中預設的整體威脅表面的非分佈數據的及時數據上進行評估。

8 結論與未來方向

本調查提供了一個全面且當前的對抗性攻擊分析，強調了網路安全威脅的動態格局，並專門關注網路入侵偵測系統。它作為研究人員、組織和網路安全專業人士的有價值資源，增強了他們對這個快速發展領域的理解。此外，本調查詳細說明了威脅者採用的各種迴避技術，以繞過NIDS，識別了現有文獻中的關鍵差距，並概述了未來研究與發展工作可以借鑒的興起挑戰與洞見。

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